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REVIEW ARTICLE

Advances in pyramid solar stills: a comprehensive review of sustainable water desalination innovations

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Abstract

This study presents a comprehensive investigation into recent advancements in pyramid solar stills (PSS), focusing on how internal and external modifications have enhanced both performance and sustainability. The research critically examines the limitations of conventional solar stills in providing clean water and proposes innovative solutions to improve their productivity. Internal improvements like the integration of phase change materials (PCMs), Nanoparticles (e.g., TiO_2 and CNT-water Nanofluids), and energy storage materials (e.g., paraffin wax and quartz rock), meaningfully improve desalination output. PCM integration alone enhances water productivity by 35 to 101.5%, while Nanoparticle application assures an efficiency gains ranging between 6.1 to 54.4%. External modifications such as the integration of solar collectors, reflectors, and forced condensation systems, has increased water productivity. Statistically, the with water yield increases to 194% with a thermal efficiency up to 62.4%. Hybrid systems, that integrate multiple modifications, establish the greatest performance enhancements, delivering up to a 166% productivity growth when PCMs and reflectors are utilised in tandem. The results highlight that optimised PSS, developed through multidisciplinary approaches, offer a potential, sustainable, and cost-effective solution for freshwater production. However, a number of barriers linked to component integration and large-scale applications remain. More importantly, the associated findings of this review have stated a foundational framework to advance the design and operation of solar desalination technologies.

Keywords Pyramid (triangular) solar still (PSS) · Internal and external modifications · Desalination · Sustainable water solutions design

List of symbols

SS	Solar still
ANN	Artificial neural network
CB	Carbon black (nanoparticle)
CBSSS	Conventional basin solar still system
CFD	Computational fluid dynamics
CNT	Carbon nanotube
CPL	Cost per liter
CPSS	Conventional pyramid solar still
CRPSS	Circular pyramid solar still
CSS	Conventional solar still
DSSS	Double-slope solar stills
ESMs	Energy storage material
FR	Filling ratio
FSS	Floating solar still
HCPF	Hollow cylindrical perforated fin
HE	Heat exchanger
HNF	Hybrid nanofluid
HPSC	Heat pipe solar collector
HTC	Heat transfer coefficient
HTR	Heat transfer rate
IRPF	Inclined rectangular perforated fin
LCF	Light cotton fabric
LHS	Latent heat storage
LHSM	Latent heat storage material
LHTESS	Latent heat thermal energy storage system

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Extended author information available on the last page of the article

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MPD	Modified pyramid distiller
MPSD	Modified pyramid solar distillers
MPDSS	Modified pyramid distiller solar still
MPSS	Modified pyramid solar still
MSS	Modified solar still
N	Buoyancy ratio
PCM	Phase change material
PDSS	Pyramid distiller solar stills
PHP	Pulsating heat pipe
PSBSD	Pyramid stepped basin solar distiller
PSD	Pyramid solar distillers
PSS	Pyramid solar still
PSSCAP	Pyramid solar still with a corrugated absorber plate
PV	Solar photovoltaic
PVT	Photovoltaic thermal
Ra	Rayleigh number
RH	Relative humidity
RSM	Response surface methodology
SC	Solar collector
SPD	Solar parabolic dish
SPSS	Stepped pyramid solar still
TA	Trapezoidal absorber
TBSSS	Triangular basin solar still system
TES	Thermal energy storage
TSA	Triangular-shaped absorber
TSD	Triangular solar distiller
TSS	Triangular solar still
TPSS	Triangular pyramid solar still
VCAP	V-corrugated absorber plate
WPSS	Wick-type modified pyramid solar still

Introduction

The shortage of drinking water has become a foremost global challenge in the twenty-first century. The growing factors of population growth, industrialization, climate change, and pollution are the main challenges. This in turn has introduced the necessity of upgrading the traditional freshwater technologies to be transferred into sustainable desalination technologies (Al-Obaidi et al. 2022; Meng et al. 2025; Ehsan kianfar 2025). In this aspect, the solar stills (SSs) have been represented as a potential technology due to their low cost, simplicity, and practical design if compared to other water desalination technologies (Essa 2022). Specifically, the attention into pyramid or triangular solar still (PSS or TSS) has been increased as a result to its geometric benefits with a high-possibility to enhance the overall performance. Indeed, the recent research has demonstrated the solar distillation as the most successful water desalination technique as it relies on solar energy to evaporate and condense water. The PSS can deliver three chief merits beyond the traditional conventional basin-type stills. First of all, the condensate collection area has been improved due to increasing its triangular geometry and sloped surfaces, which directly enhances the capture of solar radiation at enhanced thermal efficacy (Hussein et al. 2024). However, the concern of high energy requirement still represents the most challenge of basic PSS design still in a comparison to other traditional desalination technologies. Consequently, continued research to invent possible improvements methodologies can be noticed. Specifically, the current research intends to conducting a number of internal and external modifications of the basic design of PSS technology. In this aspect, these modifications have included the basin design, absorber materials, and the possible utilisation of thermal energy storage (TES) media. A notable increase of water productivity was reported by Abdullah et al. (2020a) who used tray-based designs to boost the evaporation rate while applying to pyramid configurations. Also, a clear performance enhancement was declared by Essa et al. (2020) who incorporated Al_2O_3/paraffin wax composites as wall-suspended trays besides using the vapor suction mechanisms. On the other hand, the external modifications of PSS technology were employed by a number of scientists to improve the overall performance. These were included the integration of reflectors as auxiliary systems, and improved condensation mechanisms. For instance, porous breathable belts and Nanofluids were used by Abdullah et al. (2023a) to improve the overall water productivity in rotating wick solar stills. Also, the combination of convex and dish absorbers with specialized wick materials was introduced by Omara et al. (2022), which causes a remarkable improvement of productivity of PSSs. Moreover, the concept of integrated PCMs has been introduced as a feasible solution to improve the water productivity of PSS technology. Specifically, the PCMs work as thermal energy storage during peak solar radiation periods with releasing it during off-peak times, thus outspreading the operating hours of the still. In this respect, the enhancement of PSS using PCM was introduced by Farhan Lafta Rashid (2019) and Abdullah et al. (2020b) who integrated reflector and Nanoparticle doping. Interesting, boosting the performance of PSS using nanomaterials has been appeared as a potential solution. For instance, hybrid Nanofluids (HNFs) were integrated in PSS energy systems by Sajjad Firas Abdulameer (2024). The researchers emphasized the capacity to enhance thermal characteristics as a consequence to this integration. Furthermore, TiO_2 Nano-layers were used by Shanmugan et al. (2020) to enhance the overall performance of single-basin SSs, which paved the way towards its useful use in pyramid configurations. The response surface methodology (RSM) was then introduced by Farouk et al. (2022) to model and optimise PSS performance with different types of

Nanoparticles. Mohammed et al. (2022a) examined hollow cylindrical perforated fins (HCPFs) and inclined rectangular perforated fins (IRPFs) as performance-boosting features in PSSs. The inclusion of baffles, reflectors, and vapor extraction systems was introduced by Essa et al. (2023) to improve the thermo-enviro-economic performance of cord pyramid distillers. It was stated that such modifications can essentially attain the optimum operational competence.

Recent investigations have discovered the coupling of PSSs with external energy systems. Shareef et al. (2020) increased SS output by incorporating solar collectors (SCs) with PCMs. Abdullah et al. (2023b) investigated a single-slope SS integrated with copper heating coils, internal and external reflectors, and an external condenser paired with PCM. Hybrid systems combining PSSs with other desalination technologies are also gaining momentum. Elhenawy et al. (2023) developed a membrane distillation/humidification-dehumidification system coupled with a vapor extraction fan. Elamy et al. (2024a) introduced a cylindrical SS incorporating a Nano-enhanced PCM, a vapor extraction fan, and parabolic solar concentrators. These integrated systems combined multiple mechanisms to overcome limitations inherent to single technologies. Comprehensive performance evaluations of PSSs now include energy and exergy analyses, as well as economic and environmental assessments. Essa (2024) evaluated energy and exergy efficiencies, alongside economic and environmental metrics, in a modified spherical SS using revolving ball technology with PCM. El-Sebaey et al. (2023) conducted thermal and economic analyses on cylindrical sector and dual-slope single-basin SSs. Attia et al. (2024) improved the performance of conical solar distillers through exergo-economic optimisation using experimental evaluations involving graphite bars. Advanced optimisation techniques, including machine learning and algorithmic modeling, have been applied to improve PSS design and performance. Pavithra et al. (2022) used neural network-based optimisation to predict the output of wick-based perched solar stills. Bamasag et al. (2022) developed machine learning models for forecasting and improving dish-type solar distillers integrated with convex stepped absorbers and Nano-enhanced PCMs. Environmental sustainability in PSS development has gained increasing focus. Attia et al. (2024) investigated the performance of high-efficiency solar distillers using novel green composite thermal insulation materials. Such efforts bring into line with the dual objectives of environmental protection and progressed freshwater productivity. Several ground-breaking layouts continue to arise. Alshamrani (2023) inspected performance indicators of double-glazed solar distillers that integrated Nano-PCMs and external condensation. Abdullah et al. (2022) inspected tubular SSs utilising glass cylinder tilts, Nano-coatings, and Nano-PCMs. Essa et al. (2021) improved the efficiency of tubular SS via experimental and theoretical investigations on rotating drum systems. Floating solar stills were elaborated as an innovative technique in water desalination research. Rashid et al. (2022) analysed the performance of floating solar stills, discoursing integrated membranes, design features, and operational advantages like the thermal insulation. Although a notable enhancement was approved in this context, additional enhancements are required in operational competence, scalability, besides investigating effectual hybrid systems of water distribution. Thus, there is a necessity to continue a specific trend of research to discourse the existed challenges by developing optimised materials, configurations, and hybrid systems. Future advancement in PSS technology would expected dependent on interdisciplinary associations comprising materials science, fluid dynamics, thermal engineering, and computational modeling.

Regarding to the above-mentioned investigations, it is fair to mention that there is a deficiency of a detailed systematic review to judge the current enhancements in terms of both internal and external improvements of pyramid solar stills. Accordingly, this review intends to fill this gap in the open literature. The current review is of a degree of necessary as it focuses on delivering a comprehensive inspection of recent developments in PSS, specifically emphasizing on both internal and external improvements that have expressively upgraded the basic performance and sustainability. Introducing a serious analysis of related adjustments such as the incorporation of PCMs, Nanoparticles, and enhanced energy storage materials, this review would underscore how these discoveries can overcome essential challenges and lead to substantial enhancement in water desalination systems. This review collects and synthesizes both experimental and theoretical research to offer appreciated visions for experts, engineers, and specialists working toward efficient solar desalination systems to global water shortage. In a summary, this review has been directed to aid the global objectives to present sustainable water management techniques and safe drinking water while lessening environmental effects.

Pyramid solar still with internal and external modifications

The current section focuses on revising the newest internal and external modifications in PSS technology, which intended to optimise the overall performance and sustainability. The internal modifications of integrated Nanoparticles, energy storage materials (ESMs), and PCMs have been indicated to meaningfully improve the water productivity and competence. This section also assesses the utilization of

external modifications of system components, such as solar collectors, reflectors, and forced condensation systems. These medications were enabled to improve the operational performance. Furthermore, this section would synthesize the associated results from multiple investigations to signify the advantages of integrated different internal and external modifications into hybrid desalination systems to assure the optimum performance. Last but not least, this section deliberates how optimum solar still designs can address the shortcomings of traditional distillation technologies to deliver more economical and sustainable freshwater production.

Internal modifications

Riahi et al. (2013) built four separate triangular solar distillation systems to purify lake water. In this respect, two of these systems were contained black sponges and sand layers in the basin, whereas another functioned as a conventional solar still (CSS). A 50 W solar photovoltaic DC heater was also integrated into one of the modified designs. This research was intentionally designed to improve the freshwater production in the tropical climate of Malaysia. The findings demonstrated that the layouts with the solar panel–heater and sand layer can attain increases of water productivity of 43% and 33%, respectively, in a comparison to the conventional solar still.

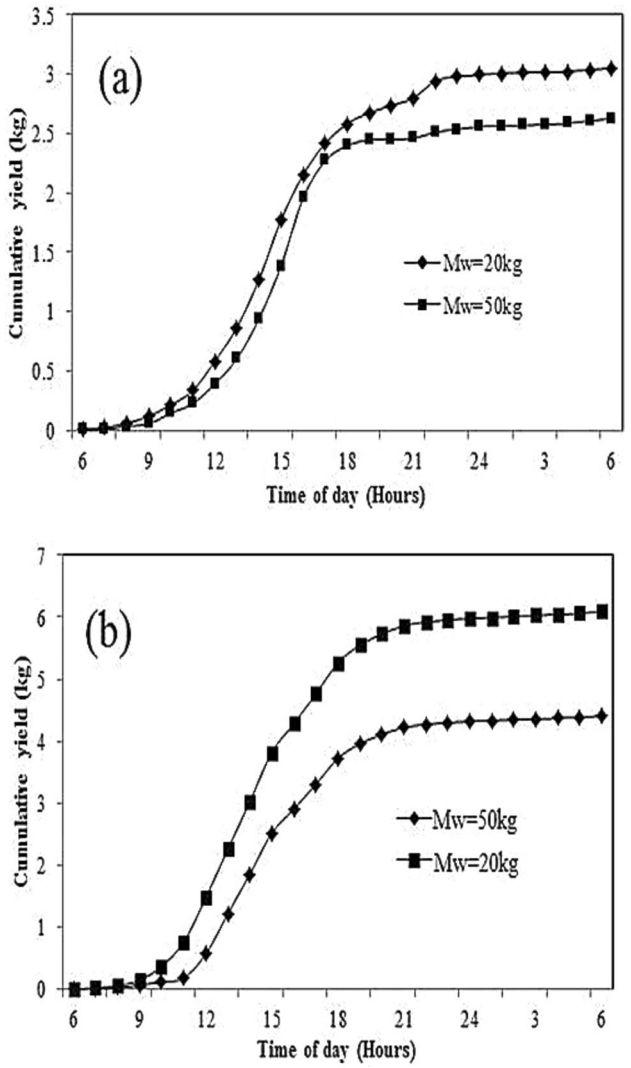


Fig. 1 The SSs total yield: (**a**) with PCM and (**b**) without PCM (Sathyamurthy et al. 2014a)

Sathyamurthy et al. (2014a) explored the impact of water mass on the competence of PSS with and without latent heat storage. Under the hot and humid climatic conditions of Chennai, India, the researchers intended to contrast the water productivity between systems prepared with and without a latent heat thermal energy storage system (LHTESS). The results showed that water productivity can be raised due to decreasing the water mass. Figure 1 shows that the inclusion of LHTESS can roughly lead to a 35% growth in water productivity in a comparison to the system without it. Also, the water productivity was greater for larger water masses throughout off-sunshine hours. Statistically, 5.5 L/m^2 day was assured for the system with LHTESS compared to 3.5 L/m^2 day for the system without it.

Under the hot and humid climatic conditions of Chennai, India, Sathyamurthy et al. (2014) presented a specific technique which involves two layouts both with and without LHS to upgrade the overall performance of Triangular Pyramid Solar Stills (TPSSs). The researchers intended to fabricate and design solar stills while comparing the water productivity in the attendance and nonappearance of a LHTESS. The researchers used paraffin wax as the LHTESS material as a result to its practical, low-cost, and extensively appropriate features. The experimental work presented that the hourly water productivity can be marginally increased in TPSSs without LHTESS on sunny days. However, the water productivity of systems incorporating LHTESS was roughly 35% higher in a comparison to stills without thermal storage. Furthermore, the researchers also stated that water productivity is particularly greater in the LHTESS-equipped stills during periods without sunlight. In this regard, the still with LHTESS produced 4.5 L/m^2 day during the summer, if compared to the still without it, which yields 3.5 L/m^2 day. However, the configurations were produced 3.4 L/m^2 day and 2.3 L/m^2 day, respectively, in winter. The inclusions of a PCM into the solar still system has attained a 20% growth in water productivity (Fig. 2).

Under similar environmental conditions in Tanta City, Egypt, a standard PSS and a modified version integrating PCMs and a V-corrugated absorber plate (VCAP) were constructed and tested by Kabeel et al. (2017). The findings were contrasted against the conventional PSS to appraise the enhancement of the overall performance of the modified

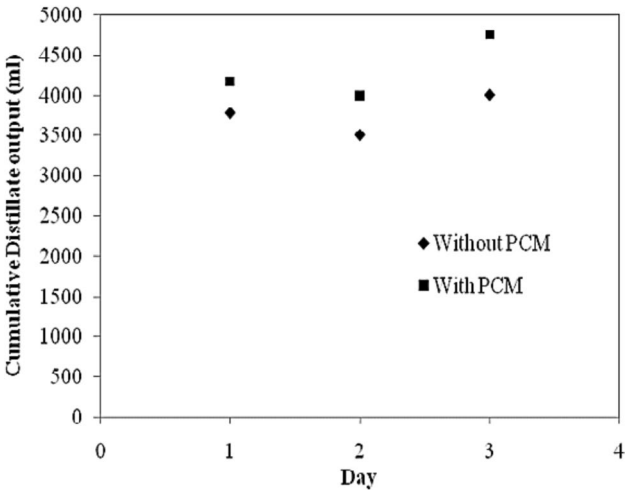


Fig. 2 Variations in cumulative yield throughout several summertime trial days (Ravishankar Sathyamurthya et al. 2014)

PSS. A greater water productivity was elucidated sing the modified PSS, that integrates PCM and VCAP beneath the basin. Statistically, in a comparison to 6.6 L/m^2 day of the modified PSS with PCM, the water productivity of standard PSS was only 3.5 L/m day. Ind^{2}eed, this is an enhancement of 87.4% in water productivity.

Kabeel et al. (2019) evacuated the competence of an innovative PSS that uses TiO_2 Nanoparticles as a coated material of the absorber plate. The experimental work was directed to consider different water depths between 1 cm to 3.5 cm while considering 0.25 m^2 as the absorber plate area. The findings indicated that the addition of TiO_2 nanoparticles (coated absorber) can rise the water temperature by approximately 1.5 °C if compared to an uncoated absorber. The temperature increase is principally understandable throughout sunlight hours, that contribute to improved efficacy. Accordingly, the water productivity of 1 cm water depth of coated absorber plate was increased to 6.6 kg/m^2 if compared to 6.2 kg/m^2 for uncoated plates.

Kabeel et al. (2020) concentrated on improving the PSS performance utilising other performance-enhancement components. In this aspect, the researchers used a PCM tank that installed under the absorber surface, while attaching a circular array of hollow copper fins to the absorber plate to improve the water productivity beyond sunset. In this research, the efficiencies of a conventional PSS, a PSS with hollow circular fins, and a PSS with both PCM and fins, were evaluated and compared while employing a similar climatic conditions. To preserve consistent basin salinity, another feed tank was supplied saline water, while maintaining water depth at 2 cm. the results indicated that the conventional PSS can achieve an extreme water productivity of 4.02 L/m^2/day. The inclusion of hollow copper fins helped to increase water productivity to 5.75 L/m^2/day, which denotes an improvement of 43%. The daily water productivity was augmented to 8.1 L/m^2 day after including the PCM. Statistically, this is an improvement of 101.5% beyond the reference layout.

Ghachem et al. (2021) presented a 3D computational research to visualize the impacts of carbon nanotube (CNT)-water Nanofluids on double-diffusive convection in TPSSs. The researchers were considered a number of effectual factors such as nanoparticle volume fraction ($0 \le \varphi \le 0.05$), Rayleigh number ($10^3 \le Ra \le 10^5$), and buoyancy ratio ($-10 \le N \le 0$). The researchers analysed differences in flow structure, temperature, and heat and mass transfer rates. The findings exposed that the inclusion of CNT would improve the thermal performance of solar stills, while the buoyancy ratio functions as a hopeful optimisation parameter for heat and mass transfer rates.

Abdallah et al. (2021) suggested numerous invented layouts and designs of single-slope solar stills. Specifically, the integration of spherical and pyramidal layouts was assured a clear improvement to the conventional pyramid design of solar stills. The effect of these design enhancements on the performance indicators of pyramid solar still was experimentally appraised. Statistically, the results indicated an increase of more than 57.1% of water productivity if compared to a conventional solar still, specifically in the semispherical and stepwise basin layouts. More importantly, the improved spherical design of pyramid solar still has necessitated lower energy to raise the vapor at reduced evaporation and condensation times in addition to reducing the distance between the water surface and the glass cover. Also, more efficient solar capture was ascertained by using the spherical geometry where the incoming radiation persists approximately perpendicular to the metal sheet and water surface from sunrise to sunset.

Prakash and Jayaprakash (2021) introduced a new solar desalination system of stepped multiple basins settled in a pyramid layout. Four basins with small surface areas were fabricated and introduced as the condensation glass cover. This design enabled to mitigate the overall thermal rate as a result to decreasing the evaporating surface area facing the condensing surface. The researchers ascertained that the suggested PSS can attain an average water productivity of 3.25 L/day (including nighttime production) while demonstrating an overall efficacy of 50.85%.

Alawee et al. (2021) presented a novel solar stills system that incorporates wick materials. This design was characterised by installing a parallel upper basin liner, positioned 3 cm above the original solar stills basin liner, and perforated it with a series of openings. The upper liner was coated with wick material and the original basin liner was filled with salt water to 2.5 cm of water depth. Also, the wick cords were suspended from the perforations and corners

of the upper liner to draw salt water upward and preserve a continuously moist wick surface. The developed design maximized the overall performance of wick usage and avoids thermal energy loss by guaranteeing the drowning up of equivalent salt water to the evaporated volume without adding or removing hot water, i.e., minimum thermal waste. The researchers explored the impacts of two wick materials (jute and cotton) and different numbers of wick cords (9, 16, 25, and 35). The findings indicated that the thermal performance of corded wick solar stills can meaningfully outpaced the conventional solar stills. Furthermore, the use of jute wicks solar stills introduced consistent operation compared to those designs that use cotton.

Using a 3D numerical methodology, Ghachem et al. (2021) inspected the effect of CNT–water Nanofluids on double-diffusive convection in a TPSS. A broad range of controlling parameters were analysed, including Rayleigh number ($10^3 \leq Ra \leq 10^5$), buoyancy ratio ($-10 \leq N \leq 0$), and nanoparticle volume fraction ($0 \leq \varphi \leq 0.05$). The researchers appraised the changes in flow structure, temperature, heat and mass transfer rates. The findings showed that the integration of CNTs would meaningfully improve the efficiency of the solar still. In this regard, it was ascertained that the buoyancy ratio has a vital role to optimise the heat and mass transfer rates.

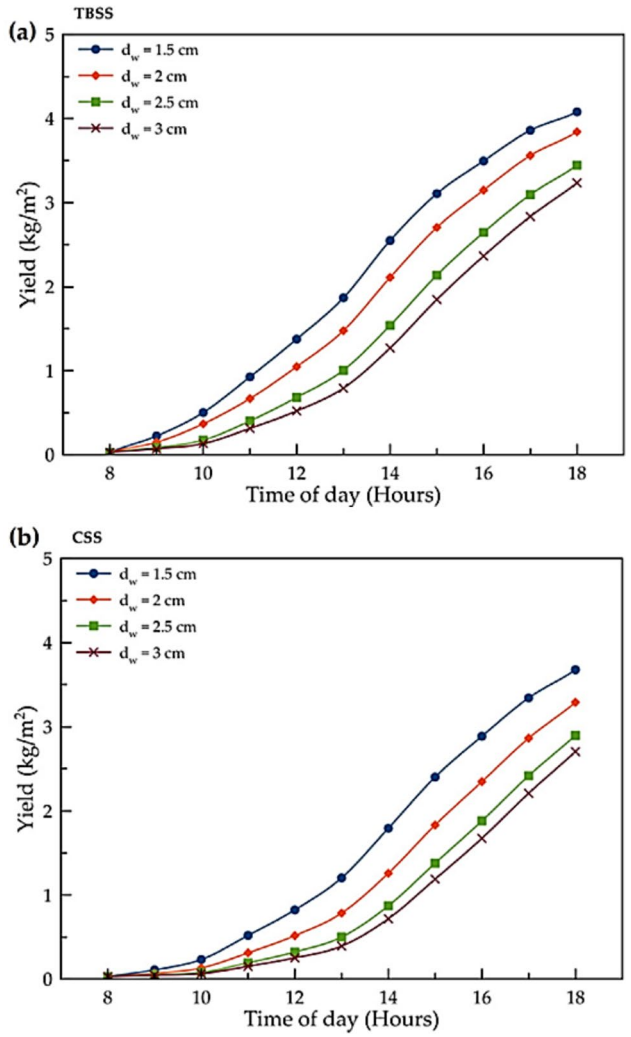


Fig. 3 Water productivity against operational time of (**a**) TBSSS and (**b**) CBSSS at different water depths (Sundeep 2022)

Modi and Shah (2022) improved the efficiency of TPSS by integrating triangular cross-sectional aluminum hollow fins and wool fabric wick fins. The researchers conducted a set of experiments of TPSS, one set with and without fins, and the other with partially and fully submerged fins. The hollow fins design of TPPS was indicated to attain total daily water productivity of 2.5674 L/m^2 with an increase of 12.36%. While, the wick fins attained 2.5481 L/m^2 of water productivity with an increase of 11.51%. Also, the water productivity of TPSS without fins was 2.2850 L/m^2. These figures include full-day operation. For partially immersed fins, wick-fin-equipped stills produced more distillate in the morning (09:00–13:00), while hollow-fin-equipped stills yielded more in the afternoon (14:00–17:00). For the case of fully submerged fins, total daily distillate outputs (including nighttime production) were 2.0668 L/m^2 without fins, 2.2529 L/m^2 with hollow fins, and 2.3042 L/m^2 with wick fins, representing increases of 9.00 and 11.49%, respectively. Efficiency gains with partially immersed fins were 15.70% for wick fins and 13.12% for hollow fins. With fully submerged fins, efficiency increased by 17.36% and 29.34%, respectively.

Siddula et al. (2022) introduced an experimental work to investigate the consequence of water mass on the water productivity of solar stills with triangular slopes and traditional single slopes. The polyethylene was used as the condensing cover for triangular slope solar stills, with 0.25 m^2 of surface area of solar stills system. The experiments considered the changes of several parameters of the solar intensity, wind speed, and ambient temperature. In other words, the researchers evaluated the impacts of variable water mass on the performance of both developed triangular slope solar stills designs. The results assured that the performance indicators of triangular single-slope solar stills outperformed the traditional counterparts. The triangular slope solar still basin achieved 6.8 °C of water temperature, that is higher than the traditional design. Also, the square basin attained water productivity of 3.29 kg/m^2 compared to the triangular basin of 3.67 kg/m^2. Figure 3 depicts that conventional basin solar still system (CSS) has assured a water productivity of 3.67 kg/m^2 for a water depth of 1.5 cm while the triangular basin solar still system (TBSS) had an enhancement of 7.08% of water productivity.

Kumar et al. (2022) directed an investigation to assess the effects of different energy storage materials on the

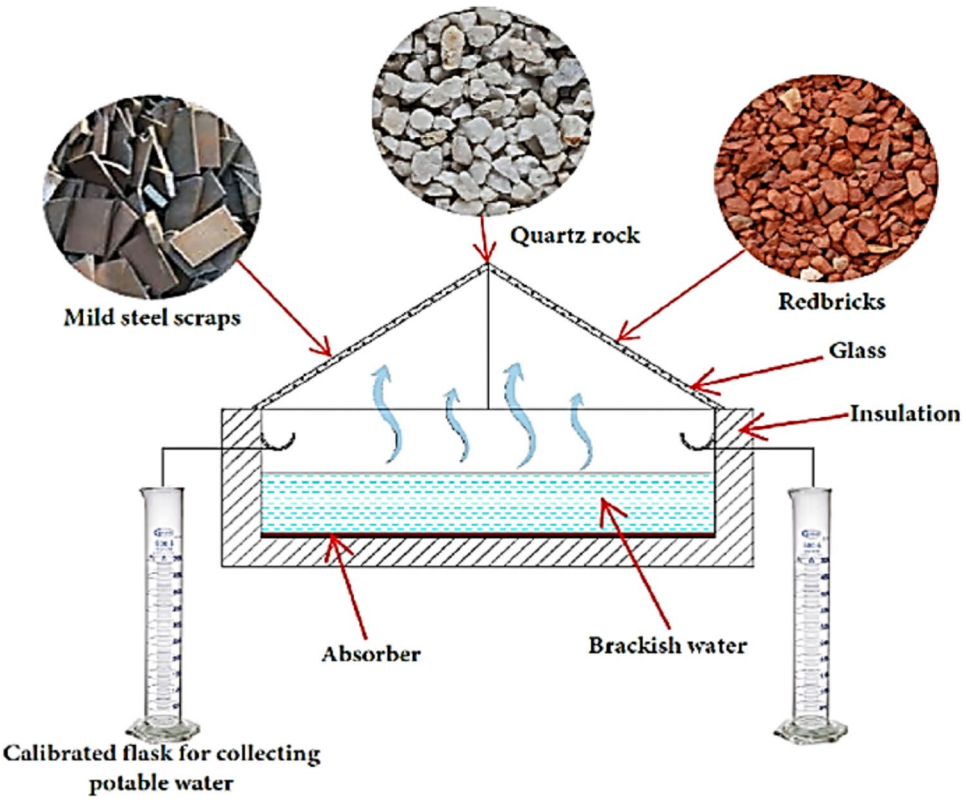


Fig. 4 PSS schematic diagram with the energy storage (Senthil et al. 2022)

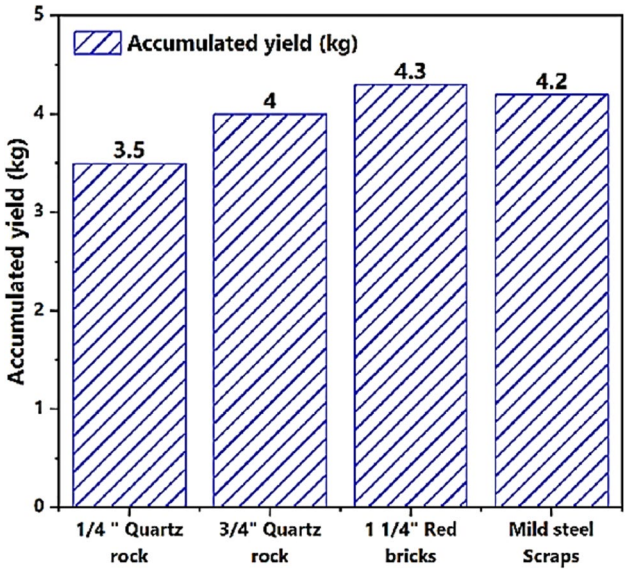


Fig. 5 Using various energy storage materials, solar stills may produce fresh water cumulatively (Senthil et al. 2022)

performance of a TPSS operation at a fixed water level (Fig. 4). To evaluate the effectiveness of various TES substances, experiments were conducted using a PSS. The basin of the solar still was constructed with different materials, including mild steel, sieved red bricks, and quartz rock, to enhance the quantity of freshwater produced. To calculate the evaporative and convective heat transfer coefficients (HTCs) from the experimental data, conventional correlation methods were employed. The use of TES materials in the basin increased freshwater production by approximately 30% compared to a standard solar still. This improvement was achieved by incorporating 1¼-inch red bricks in place of conventional materials. Furthermore, as illustrated in Fig. 5, the daily freshwater yields for different materials were as follows: 3.5 L/day for 1¼-inch quartz rock, 4 L/day for ¾-inch quartz rock, 4.3 L/day for 1¼-inch red bricks, and 4.2 L/day for mild steel waste.

Modi and Shah (2022) reported that integrating triangular cross-sectional aluminum hollow fins with wool fabric wick fins significantly improved the efficiency of TPSS. Two sets of tests were conducted to evaluate TPSS performance: one comparing systems with and without fins, and the other comparing partially and fully immersed fins. With hollow fins, the total distillate output per square meter was 2.5674 L/m^2 (a 12.36% increase), while with wick fins, it was 2.5481 L/m^2 (an 11.51% increase). The system without fins produced 2.2850 L/m^2. These values represent daily totals, including full-day operation. In the case of partially immersed fins, wick-finned stills produced more water in the morning hours (09:00–13:00), while hollow-finned stills outperformed in the afternoon (14:00–17:00). The total daily water productivity (including nighttime production) for fully immersed fins were 2.0668 L/m^2 without fins, 2.2529 L/m^2 with hollow fins, and 2.3042 L/m^2 with

wick fins, demonstrating growths of 9.00 and 11.49%, respectively. Statistically, the gain of efficacy with partially immersed fins was 15.70% for wick fins and 13.12% for hollow fins. However, the efficacy can be improved with completely submerged fins, that obtains 17.36% for hollow fins and 29.34% for wick fins.

The performance of a PSS was enhanced by (Ghandourah et al. 2022) via enlarging the evaporation surface via the use of a corrugated absorber plate (PSSCAP). The overall performance of PSSCAP was appraised under the climatic conditions of Patan, Gujarat, India, together with a CSS. The findings indicated that the PSSCAP can generate greater water productivity with the increase in water temperature. Particularly, the water productivity of PSSCAP was 4.5 L/day, in a comparison to 2.95 L/day for the CSS. This is an improvement of 52.54% of water productivity that can be attributed to the corrugated absorber plate (Fig. 6). Akso, the the PSSCAP attained an average thermal efficiency of 45.5% compared to 31.5% for the CSS. The assessed unit cost of water production was around 0.53 INR for the PSSCAP in a comparison to 0.68 INR for the CSS.

An experimental research was conducted by Asadabadi and Sheikholeslami (2022) to delve into the impacts of insulation and copper (Cu) fins on the efficiency of solar stills system. Three scenarios were designed in this research under three diverse test conditions while comparing the performance of a TPSS unit against a MPSS unit. These developed layouts permitted to test the impacts of variable fin numbers and insulation levels. The findings indicated that water productivity can be augmented to 62.5% using the MPSS unit with attaining an optimum daily water productivity of 5.33 L/m^2. Moreover, the depicted water daily productivity of the first, second, and third MPSS layouts were 39.64, 43.3, and 48.12%, respectively. The cost analysis indicated the cost-benefits of the developed first, second, and third MPSS layouts. Statistically, these layouts came with a reduced cost of freshwater by 12.76, 16.68, and 27.33%, respectively, with the use of glass wool insulation and hollow circular Cu fins.

Mohammed et al. (2022b) addressed the efficiency of a circular pyramid solar still (CRPSS) upgraded with two types of fins, inclined perforated rectangular fins (IPRF) and HCPF, in an integration to nanocomposites of TiO_2 and graphene. The upgraded systems, denoted to PSS-IPRF and PSS-HCPF, established meaningfully better water productivity compared to the base CRPSS. PSS-IPRF and PSS-HCPF demonstrated a daily water productivity of 4840 mL/m^2 and 3718 mL/m^2, respectively, that signify an increase of water productivity of 55.9% and 31.3%, respectively. Also, the inclusion of nanocomposites can further enhanced the daily water productivity of PSS-IPRF by 82.1%.

Sudhakar et al. (2023) discovered the impacts of integrating a latent heat storage material (LHSM), explicitly paraffin wax, into the absorber base of both double-slope and TPSS units. The researchers contrasted the efficiency of solar stills with and without LHSM under classic climate conditions in Chennai, Tamil Nadu, utilising a fixed water depth of 30 mm. The results clearly showed that paraffin-based LHSM can meaningfully the overall efficiency, with the most considerable enhancement detected in the TPSS layout.

Ahmed et al. (2023) investigated how cylindrical fins of variable water heights, encapsulated with PCM can enhance the efficiency of PSS. Three similar PSS units were tested, each with a 1 m^2 basin. In Phase 1, conventional stills were compared with those equipped with 40 mm-high fins, under various water depths (10–40 mm). Phase 2 investigated the influence of different fin heights (20, 40, and 60 mm), and Phase 3 examined the impact of encapsulating 40 and 60 mm fins with paraffin wax. Phase 1 results showed optimal performance at a water depth of 10 mm with fins. Phase 2 findings revealed that 40 mm fins delivered the highest overall efficiency at 43.4%, compared to 38.5% and 35.7% for 20 and 60 mm fins, respectively. In Phase 3,

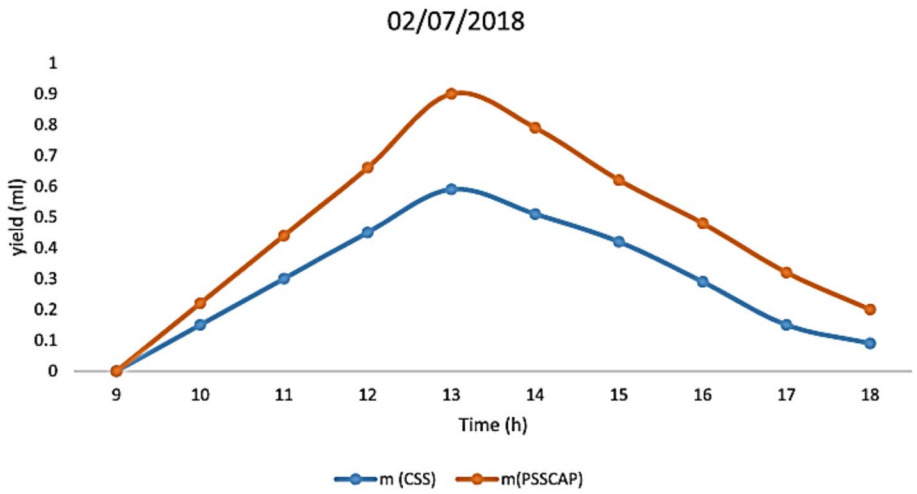


Fig. 6 A comparison of water productivity between CSS and PSSCAP (Ghandourah et al. 2022)

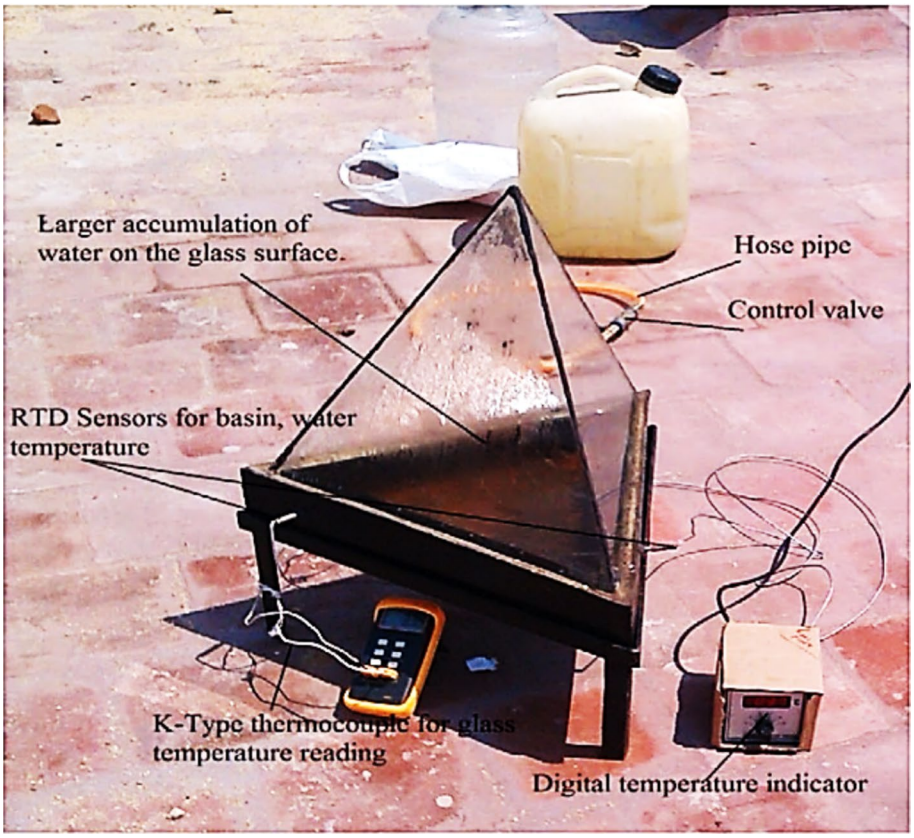


Fig. 7 Triangular pyramid solar still (Sathyamurthy et al. 2014b)

encapsulating 40 mm fins with PCM yielded an overall efficiency of 49.9%, compared to 34.6% for traditional stills, an improvement of approximately 44.4% in water production.

Shanmugan et al. (2023) explored the thermodynamic performance and chemical potential connections of condensed water droplets and vapor within a pyramid stepped basin solar distiller (PSBSD). The results indicated that the PSBSD's internal heat transfer coefficients are affected by ambient temperature and solar radiation intensity. The chemical potential analysis stated how liquid–vapor equilibrium and temperature changes across components can impact the system performance. The PSBSD achieved a daily water productivity of 4.28 L/m^2 at an efficacy of 38.135% over a 24-h cycle.

Chauhan et al. (2023) discoursed the effect of incorporated thermal storage of paraffin wax with porous clay pots on the efficacy of a PSS technology. Clay pots, either unpainted or painted black are placed upside down on the absorber plate to improve the evaporation rate. Particularly, three layouts were utilised. These include the standard PSS without clay pots or PCM, a PSS with 25 unpainted clay pots and paraffin wax, and a PSS with 25 black-painted clay pots and paraffin wax. The results indicated the water productivity of 2.21 L/m^2, 3.20 L/m^2, and 3.63 L/m^2 for the three layouts, respectively.

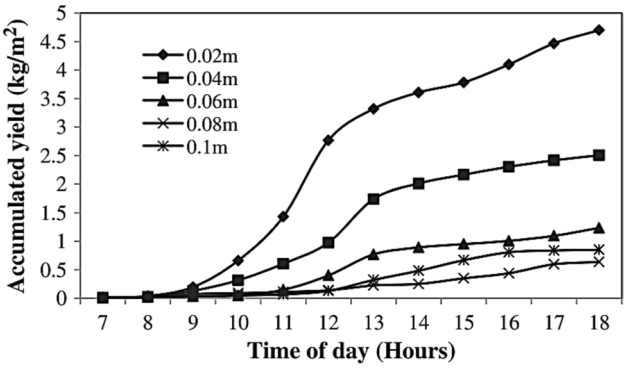


Fig. 8 Distillate rate fluctuations during the day at varying water depths (Sathyamurthy et al. 2014b)

Sathyamurthy et al. (2014b) investigated the efficiency of a number of effectual parameters that control the overall efficiency of TPSS system while outlining the impacts of water depth and its consequence on the evaporative and condensation heat transfer coefficients as depicted in Fig. 7. The researchers highlighted that an optimisation of solar distillation necessitates careful control of the temperature difference between the condensation and evaporation surfaces. Notably, TPSSs typically have a larger condensing surface area than evaporating area. As shown in Fig. 8, the maximum freshwater yields were 4.3 kg/m^2 at a depth of 0.02 m, 1.2 kg/m^2 at 0.04 m, 0.9 kg/m^2 at 0.08 m, and 0.5 kg/m^2 at

0.1 m. Additional outdoor tests demonstrated that increasing wind speed from 1.5 m/s to 3 m/s and 4.5 m/s enhanced productivity by 8.5% and 15.5%, respectively, due to improved glass cover cooling.

The thermal performance of a Triangular Solar Distiller (TSD) was investigated by Boudhiaf et al. (2024) using a novel Trapezoidal Absorber (TA) design. In the redesigned system, referred to as TSDTA, the shadow cast by the sloped sidewalls and the water volume is minimized. Under identical climatic conditions in Sfax, a city in Central-Eastern Tunisia, the performance of the TSDTA was compared experimentally with a traditional TSD equipped with a flat Plate Absorber (TSDPA). The findings showed that the trapezoidal absorber can enhance the diurnal water productivity by 22.73% in a comparison to the plate absorber. In this aspect, the thermal competence of the TSDPA was 46% during daytime operation, while the TSDTA attained a greater efficacy of 49%. Figure 9 indicated that the exergy efficiency can be augmented throughout the day, peaking at 4.62% for the TSDTA and 4.37% for the TSDPA at 2:00 PM—probable as a result to improved water productivity during peak sunlight hours.

Zakaria et al. (2024) examined the efficiency of TPSSs that use large surface areas on both sides to improve the overall heat retention. The droplet formation was promoted for the angled sidewalls and temperature gradient between the interior and exterior of the TPSSs, therefore helping water productivity. 100 mL of PCM was applied in a galvanized steel tray during a 4 to 5-h experiment. The utilisation of sugarcane bagasse as a porous medium has enhanced the overall efficiency by 2%. Also, two Peltier-based thermoelectric components augmented the internal humidity by 5%, meaningfully advancing the water productivity.

Abed et al. (2024) developed a PSS incorporating PCM and Nano PCM (containing 3% nanoparticles) and employed numerical simulations to evaluate performance. Comparisons were made between conventional SSs and those fitted with various corrugated absorber shapes (flat, triangular, quadratic, and rectangular). The findings indicated that rectangular corrugated absorbers with PCM and Nano PCM can

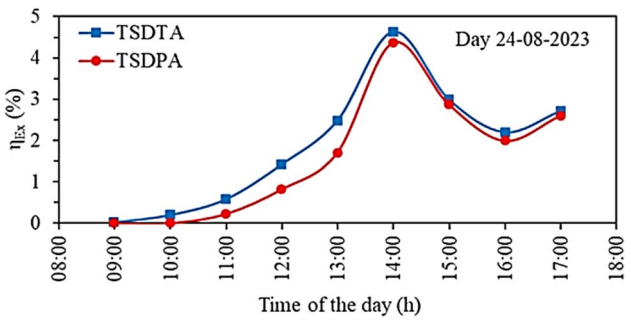


Fig. 9 Variation of exergy efficacy against the operational time for TSDPA and TSDTA (Boudhiaf et al. 2024)

improve the water productivity by 116.64% and 79.51%, respectively, beyond the traditional pyramid stills. The MPSS stated a growth of water productivity of 112.34%. For the Nano PCM-coated rectangular layout, considerable enhancements of 41.4 and 54.4% were achieved for convective and evaporative heat transfer coefficients, respectively.

The effect of rotational motion on water productivity of pyramid solar stills was elucidated by Baiee and Asiaei (2024) who inspected the overall evaporation, condensation, heat transfer coefficients. Also, the effects of shading, low water temperatures under saline conditions, and energy loss reduction were investigated. The experiments were conducted using different rotational speeds (0.125, 0.25, 1.0, and 1.5 rpm), while the pyramid solar stills operated from 9:00 AM to 4:00 PM. The findings elaborated an obvious growth in water productivity as a result to an increase in the rotation speed, which deduced an optimum efficiency at 1 rpm. It was stated that the turbulent water motion can enhance the heat transfer from the absorber plate, which hits a peak temperature of 79.1 °C at 1.5 rpm (2:00 PM). However, the absorber temperature was dropped to 78 °C in the case of not rotated system. More importantly, the water productivity was increased to 30.2% while using a rotation of 1 rpm in a comparison to the standard layout.

Dhaoui et al. (2024) investigated the effect of angled fin layouts on energy and exergy competences in both finned PSS and double-slope solar stills (DSSS) systems. In the experimental work, the researchers used different layouts of angled fins (3 × 3, 3 × 5, and 3 × 7) over multiple days. The optimum findings were attained using a finned systems PSS absorber plate of 7 × 3 fins, that characterises by augmented water productivity by 43.22% in the PSS and 46.25% in the DSSS in a comparison to un-finned absorbers. 7 × 3 finned PSS entailed an energy efficiency of 37.84% in a comparison to 31.93% for the DSSS. In this aspect, the energy efficiency was reduced to 26.18% for the PSS and 22.39% for the DSSS for the scenario of without fins.

Essa et al. (2024) focused on enhancing the thermal performance of PSD by intoducing a modified version (MPSD) with three vertical walls supporting 4 cm-wide trays. The system was designed to generate variable speed dragging fan (1.0–3.5 rpm) connected to an external condenser in addition to test the system with and without a fogging mechanism. The MPSD with suspended trays produced 35% more distillate than the unmodified PSD. The fogging system enabled to increase the water productivity 160%, while attaining optimal thermal performance at 2 rpm. More specifically, the MPSD system attained a considerable thermal performance of 62.4% and an increase of water productivity of 194%. The developed technology approved the reduction of water production cost from $0.24/L (PSD) to $0.18/L

(MPSD), making this technique as the most promising for effective solar desalination.

Anika et al. (2024) evaluated the performance of four upgraded scenarios of CSS configurations in Rajshahi, Bangladesh. These included a baseline CSS (Scenario I); addition of fins and black sand (Scenario II); incorporation of fins, hybrid Nano PCM, black sand, and sponges (Scenario III); and fins, hybrid Nano PCM, crushed stone, and sponges represented Case IV. Padma Scenario III attained the greatest thermal performance, outstripping Scenario II by 32.44% and Scenario IV by 6.47%. Scenario entailed exergy efficiency enhancements over Scenario I were 18.89% for Scenario II, 74.75% for Scenario III, and 53.54% for Scenario IV. Payback periods for Scenarios I–IV were 185, 159, 126, and 135 days, respectively. Also, the water production cost was decreased by 73.5% (Scenario I), 77.29% (Scenario II), 81.89% (Scenario III), and 80.63% (Scenario IV) in a comparison to market rates. The energy production factors for Scenarios were I–IV were 0.137, 0.099, 0.138, and 0.195 yr^{-1}, respectively.

Diab et al. (2025) addressed the advantages of improving the solar still performance by reforming the basin into an inverted aluminum pyramid. The developed design delivered the highest active surface area at enhanced thermal storage as a result to aluminum's high thermal conductivity and solar absorption features. A significant gain if water productivity was assured by conducting the experimental and theoretical investigations that compared the performance of modified solar still (MSS) and conventional CSS. At 30 L of saline water input, the MSS achieved a 46.67% increase in productivity (4.40 kg/m^2/day), compared to a 29% improvement from the CSS. Exergy efficiency rose from 2.0% (CSS) to 3.12% (MSS). The energy payback period dropped from 17.18 years (CSS) to 7.2 years (MSS). Annual CO_2 emissions were reduced by 13.668 tons (CSS) and 20.05 tons (MSS). The cost of freshwater production was also 45.45% lower with the MSS. As shown in Fig. 10, freshwater output increased steadily as saline water volume rose from 5 to 30 L.

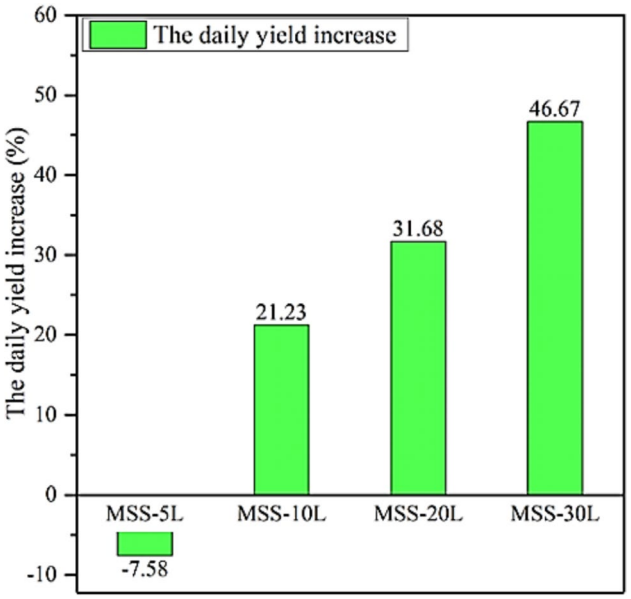


Fig. 10 The daily rise in MSS output relative to CSS at various saltwater concentrations (Mohamed Ragab Diab 2025)

Mohtasim et al. (2025) conducted an in-depth examination of hollow circular fins, sponges, black sand, and hybrid Nanoparticles with PCM in various SS configurations (Fig. 11). Additionally, the study employed 11E sustainability matrices to assess the effects of serpentine copper pipe designs integrated with hybrid Nano-enhanced PCM and

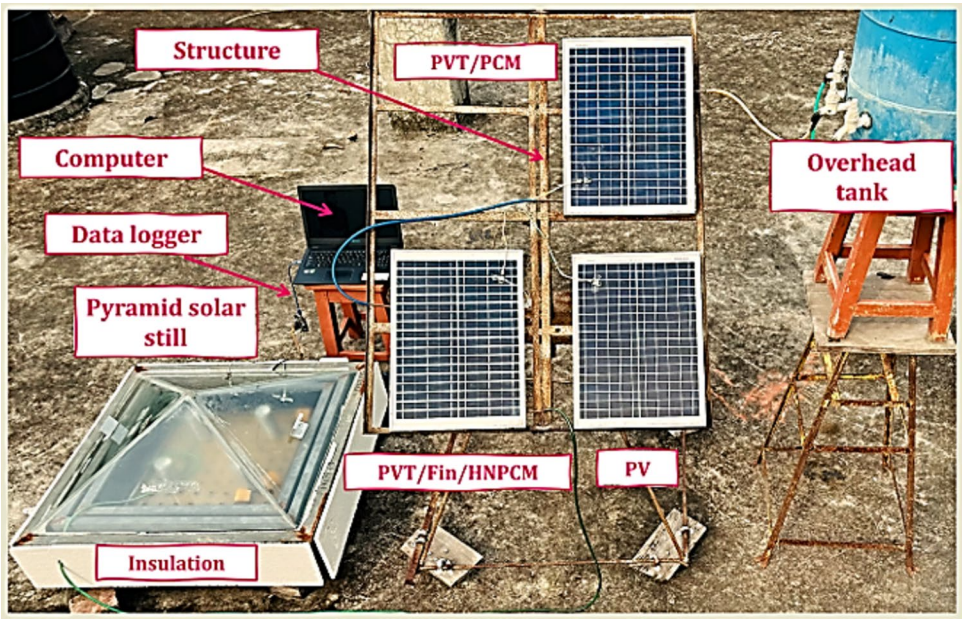


Fig. 11 An image of the whole experimental apparatus (Mohtasim et al. 2025)

fins under photovoltaic thermal (PVT) scenarios, using an average flow rate of 0.0021 kg/s. Among the evaluated configurations, the modified SS demonstrated the highest productivity, achieving a daily freshwater output of 4.109 L/m^2 and reducing cost by approximately 94.13% per liter. The PVT/HNPCM/Fin configuration significantly outperformed a conventional photovoltaic module in terms of thermal efficiency (75.11%) and electrical energy efficiency (with a relative improvement of about 15.70%). Furthermore, the adapted SS exhibited the highest exergo-enviro-economic performance, with an improvement of approximately 30.57% compared to other SS configurations. The integration of PVT with the adapted SS offered a cost-effective approach to reduce CO_2 emissions by 6.98 tons over its operational lifetime while providing substantial thermal energy for desalination. Table 1 summarises the relevant studies on TSS with internal modifications.

External modifications

In their theoretical study, (Naveen Kumar et al. 2016) incorporated a TPSS with baffles into an inclined SS. The amount of water in the basin plays a critical role in determining SS performance. Their findings showed that integrating a TPSS with baffles into an inclined SS significantly enhanced output, while analysing a range of water mass between 20 to 100 kg. based on the theoretical findings, the daytime water productivity was lessened from 46 to 6% as a result to increasing water mass. However, the nighttime water productivity was augmented from 46 to 86% as a result to TES.

A pyramid-shaped SS was built from galvanized iron sheets by Rajan et al. (2016) who tested the effects of different water depths (2 cm—4 cm), while combining a biomass heat source. A heat exchanger with five bends, a length of 3.5 m and a diameter of 0.025 m was attached to the basin's tail end. The heat exchanger transferred high-temperature thermal energy from the biomass source to maintain steady circulation in the saline water. Different materials were utilised to upgrade the basin temperature, comprising evaporative surface materials, sensible heat storage materials, and latent heat storage. In this research, three modes were evaluated including the solar-only, continuous flow, and once-through flow. Theoretical analysis using RSM software was compared against the experimental data. Sensible heat storage materials enhanced the overall performance by 48% over conventional solar stills, while evaporative materials delivered an increase of water production by 19%. More importantly, the continuous flow layout was deduced the greatest performance among the operational tested modes.

Kumar et al. (2016) ascertained the improvement of water productivity of TPSS that characterises by an integration of an inclined SS with baffles. The critical analysis assured an increase of basin water mass from 20 to 100 kg besides decreasing the daytime water productivity from 46 to 6%. Also, the nighttime water productivity was increased from 46 to 86%, which can be attributed to thermal energy storage.

An investigation of water productivity was conducted by Nagarajan et al. (2016) who employed blowing air over the entire glass surface of a TPSS. The researchers evaluated the consequence of both basin water mass and air velocity. The results showed that the optimum water productivity of 10.1 kg/m^2 can be attained at a maximum air velocity of 40 m/s, and a minimum basin water mass of 20 kg. Furthermore, the findings indicated a water productivity increase of 104% as a result to increase the wind speed from 0.5 m/s to 4 m/s, which signifies the practical relevance of wind-driven improvement in real-world conditions.

Kumar et al. (2017) presented the merit of SS integration strategies to improve the water productivity. The researchers experimentally analysed how basin water depth and flow rate affect production in a combined system of an inclined SS and a triangular PSS. In the unintegrated setup, water at depths of 0.02, 0.04, 0.06, and 0.08 m was placed in the PSS basin. In the integrated setup, water from the inclined SS outlet (at a constant flow rate of 8.33 kg/h) was directed into the triangular PSS for additional heating. Results showed that integration improved the inclined SS's production by 79.05%. As shown in Fig. 12, the greatest water temperature, and thus yield, was achieved at the shallowest depth ($d_w = 0.02$ m) in the integrated configuration.

The performance of a basin-type solar still is highly dependent on both design and operational parameters. Appadurai and Velmurugan (2017) conducted an extensive study to optimize specific design factors through experimental research, including the basin water-to-glass gap, incident solar intensity, inlet brine water temperature, and glass condensation surface area. The condensation area was upgraded by integrating a pyramidal glass cover. Also, the inlet brine water was preheated in a mini solar pond, while two external reflectors were utilised to redirect further solar radiation into the still. The stepped basin design assured a minimum gap between the glass cover and the water surface. The results indicated that incorporating a stepped basin with a mini solar pond and external reflectors can expressively improve the daily water productivity. The improved solar still can attain 4,920 mL/day of water productivity, that elaborates a growth of 66.2% if compared to a traditional solar stills.

Elgendi et al. (2022) proposed a successful design of passive pyramid solar still that incorporated an automated feed water system to compensate the loss of water evaporation. The preservation of a constant water level in the solar still is essential to upgrade the overall performance. In this

Table 1 A summary of relevant researches to TSS with internal modifications

Location	Modification	Key performance metrics	Results and remarks	Authors (year) [reference]
Perak, Malaysia	TSS with sand layer and black sponges in the basin	Productivity improvement (%)	Productivity increased by 33% (sand layer) and 43% (solar panel-heater) vs. CSS	Riahi et al. (2013)
Chennai, India	TSS with latent heat energy storage (LHTESS)	Daily yield (L/m^2 day), productivity improvement (%)	5.5 L/m2day (with LHTESS) vs. 3.5 L/m^2day (without); 35% increase	Sathyamurthy et al. (2014a)
Tamil Nadu, India	TSS with paraffin wax LHTESS	Seasonal yields (L/m^2 day)	Summer: 4.5 L/m^2day (with PCM) vs. 3.5 L/m2 day (without); Winter: 3.4 vs. 2.3 L/m2 day	Sathyamurthya et al. (2014)
Tanta University, Egypt	PSS with v-corrugated absorber plate and PCM	Accumulated distillate yield (L/m^2 day), productivity improvement (%)	6.6 L/m^2day (modified) vs. 3.5 L/m^2day (conventional); 87.4% improvement	Kabeel et al. (2017)
Tanta University, Egypt	PSS with TiO2 nanoparticle coating	Yield improvement (%), temperature increase (°C)	6.1% yield increase; water temp ↑1.5 °C with TiO2 coating	Kabeel et al. (2019)
Tanta University, Egypt	PSS with hollow fins and PCM	Daily productivity (L/m2 day), enhancement (%)	8.1 L/m2 day (PCM + fins) vs. 4.02 L/m2day (conventional); 101.5% improvement	Kabeel et al. (2020)
Riyadh, Saudi Arabia	TSS with CNT-water Nanofluid	Heat/mass transfer rates, buoyancy ratio effect	CNT improved heat/mass transfer; buoyancy ratio optimized performance	Ghachem et al. (2021)
Amman, Jordan	PSS with chamber stepwise basin	Productivity improvement (%)	57.1% increase vs. pyramid solar still	Abdallah et al. (2021)
Coimbatore, India	PSS with multiple basins	Efficiency (%), daily yield (L/day)	50.85% efficiency; 3.25 L/day yield	Prakash and Jayaprakash (2021)
Baghdad, Iraq	PSS with dangled wick cords	Thermal performance comparison	Jute wick outperformed cotton; thermal efficiency improved	Alawee et al. (2021)
Riyadh, Saudi Arabia	TSS with CNT-water Nanofluid	Heat/mass transfer enhancement	CNT enhanced heat transfer; buoyancy ratio critical for optimization	Ghachem et al. (2021)
Gujarat, India	TSS with hollow/wick fins	Efficiency improvement (%)	Partially submerged: 13.22% (hollow), 15.70% (wick); fully submerged: 17.36%, 29.34%	Modi and Shah (2022)
Telangana, India	TSS vs. single slope solar still	Yield (kg/m^2), temperature difference (%)	3.67 kg/m^2 (triangular) vs. 3.29 kg/m^2 (conventional); 7.08% yield increase	Siddula et al. (2022)
Tamil Nadu, India	TSS with energy storage materials	Yield (L/day)	4.3 L/day (red bricks), 4.2 L/day (mild steel); 30% improvement	Kumar et al. (2022)
Gujarat, India	TSS with hollow/wick fins	Efficiency improvement (%)	Same as (Prakash and Jayaprakash 2021)	Modi and Shah (2022)
Gujarat, India	PSS with corrugated absorber plate	Yield (L/day), efficiency (%)	4.5 L/day (PSSCAP) vs. 2.95 L/day (CSS); 52.54% yield increase; 45.5% efficiency	Ghandourah et al. (2022)
Shahriar, Iran	PSS with copper fins and insulation	Daily productivity (L/m2), efficiency (%)	5.33 L/m2 (max); 62.5% productivity increase; 48.12% efficiency	Asadabadi and Sheikholeslami (2022)
Baghdad, Iraq	PSS with fins and nanocomposites	Daily productivity (mL/m2), improvement (%)	5750 mL/m2 (PSS-IPRF) vs. 3718 mL/m^2 (CPSS); 55.9% improvement	Mohammed et al. (2022b)
Chennai, Tamil Nadu	TSS with latent heat storage material (LHSM)	Performance improvement	LHSM (paraffin) enhanced performance; TPy still outperformed	Sudhakar et al. (2023)
Benha University, Egypt	PSS with PCM-encapsulated fins	Efficiency (%), productivity improvement (%)	49.9% efficiency (PCM fins) vs. 34.6% (conventional); 44.4% productivity increase	Ahmed et al. (2023)
Andhra Pradesh, India	PSBSD analysis via Gibbs phase rule	Efficiency (%), yield (L/m2day)	38.135% efficiency; 4.280 L/m2day yield	Shanmugan et al. (2023)
Faridabad Haryana, India	PSS with PCM and clay pots	Productivity (L/m2), improvement (%)	3.63 L/m2 (Case 3) vs. 2.21 L/m2 (Case 1); 64.25% improvement	Chauhan et al. (2023)
Chennai, India	TSS with varying water depths	Yield (kg/m2), wind speed impact (%)	4.3 kg/m2 (0.02 m depth); 15.5% yield increase at 4.5 m/s wind	Sathyamurthy et al. (2014b)
Sfax, Tunisia	TSS with trapezoidal absorber	Diurnal production increase (%), thermal efficiency (%)	22.73% production increase; 49% max efficiency (TSDTA)	Boudhiaf et al. (2024)
Melaka, Malaysia	TPSS with sugarcane bagasse	Performance improvement (%)	2% overall improvement with bagasse; 5% humidity increase with Peltier	Zakaria et al. (2024)
Najaf, Iraq	PSS with Nano PCM and corrugated absorbers	Productivity increase (%), heat transfer coefficients (%)	117.64% productivity increase; 54.4% higher evaporative heat transfer	Abed et al. (2024)
Tehran, Iran	PSS with rotational motion	Productivity increase (%)	30.2% output increased at 1 rpm rotation	Baiee et al. (2024)

Table 1 (continued)

Location	Modification	Key performance metrics	Results and remarks	Authors (year) [reference]
Gabes, Tunisia	PSS with angled fins	Energy efficiency (%), productivity increase (%)	37.84% efficiency (PSS with fins); 43.22% productivity increase	Dhaoui et al. (2024)
Kafrelsheikh, Egypt	MPSD with trays, fogger, and condenser	Productivity increase (%), efficiency (%)	194% productivity increase; 62.4% thermal efficiency	Essa et al. (2024)
Rajshahi, Bangladesh	PSS with Nano-PCM, black sand, and sponge	Thermal efficiency (%), exergy improvement (%)	32.44% thermal efficiency; 74.75% exergy improvement vs. CSS	Anika et al. (2024)
Kafrelsheikh, Egypt	Inverted pyramid aluminum basin	Daily yield (kg/m2/day), efficiency (%)	4.40 kg/m2/day (MSS); 46.67% increase; 14% thermal efficiency improvement	Diab et al. (2025)
Rajshahi, Bangladesh	PSS with hybrid modifications	Productivity (L/m2/day), thermal efficiency (%)	4.109 L/m2/day; 75.11% thermal efficiency; 94.13% cost reduction per liter	Mohtasim et al. (2025)

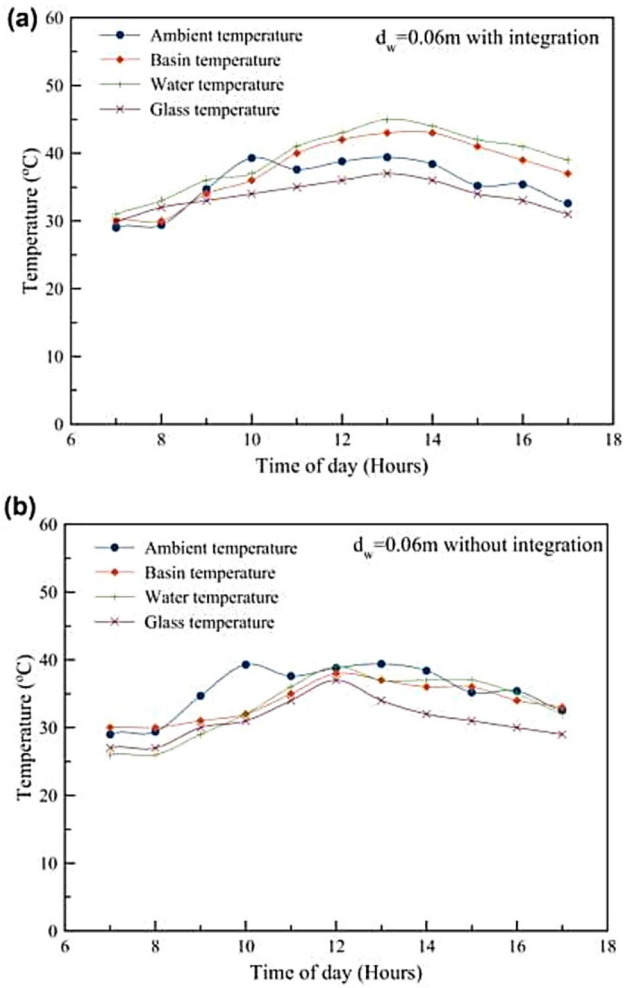


Fig. 12 Triangular PSS temperature change on an hourly basis at a fixed depth of water: **a** With and **b** Without the integration (Kumar et al. 2017)

regard, the reduction of overall efficiency is the result of reduced water level as a consequence of water evaporation. Particularly, a screw jack was incorporated to control the water level inside a supplementary tank, whereas a ball valve was connected between the solar still and the storage tank to manage the water flow inside the device. The optimum water depth within the still was assured to improve the overall performance.

A novel triangular solar distillation system developed by Ahsan et al. (2023) is depicted in Fig. 13 using locally sourced materials. The system integrates a solar photovoltaic (PV) panel connected to a direct current water heater. It utilizes solar thermal energy either directly to heat the saline water or indirectly via the water heater for the distillation process. The trough, made of black-painted Plexiglass, is mounted within a triangular UPVC frame. Experimental observations revealed diurnal variations in solar thermal energy, distillate output, temperature, and relative humidity. Linear proportional relationships were found between productivity and key parameters such as ambient temperature, water–glass temperature differential, and solar heat input. The proposed still outperformed the conventional model. As shown in Fig. 14, the two most influential factors affecting still productivity were relative humidity (RH) and ambient temperature (T_{ha}).

A new pyramid-shaped floating solar still (FSS) with decreased wettability surfaces, achieved through Si nano-coating, was proposed by Wei et al. (2024). This FSS utilizes the pyramid's central symmetry for enhanced structural stability and employs Nano-coated surfaces to improve condensation efficiency. Performance was assessed during a four-day outdoor trial. Under an accumulated solar irradiation of 15,166 MJ/m^2 day, the maximum yield from the Nano-coated FSS reached 1.017 kg/m^2 day, representing a 16.76% increase compared to the uncoated version. Furthermore, an enhancement of water productivity of 0.949 kg/m^2 day was ascertained for the Nano-coated FSS. Computational analysis of the Nano-coating's influence on droplet formation and movement revealed that condensate droplets retained their shape longer and traveled faster and farther, helping to attain a high-degree of condensation rate. The researchers conducted cost and economic analysis that deduce the economic feasibility of the proposed FSS and ascertained its viability for off-grid communities. This analysis revealed the water production cost of $0.129 per liter.

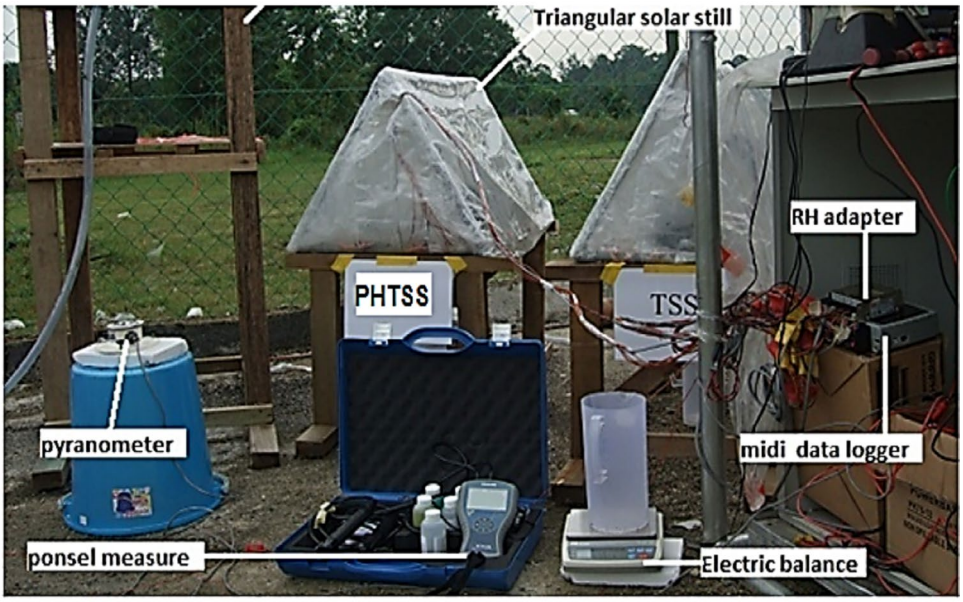


Fig. 13 Details of the PHTSS field photo (Ahsan et al. 2023)

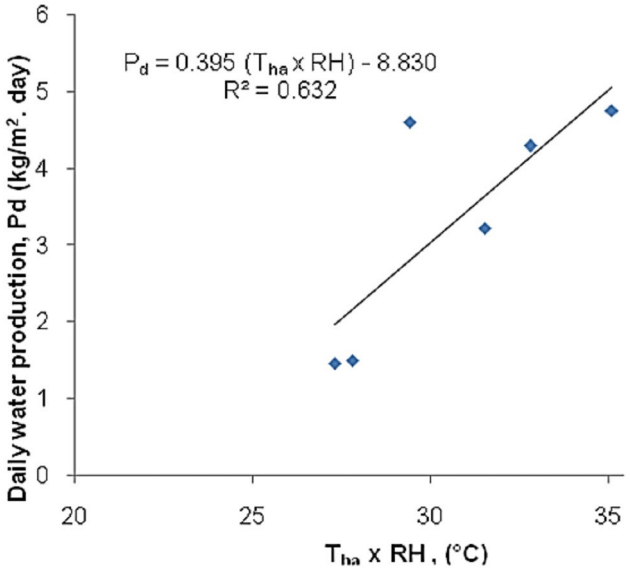


Fig. 14 Daily distillate production is correlated with the everyday mean temperature of the humid air multiplied by RH (Ahsan et al. 2023)

Pandey and Naresh (2024) investigated the feasibility of two configurations of MPSS combined to a pulsing heat pipe (PHP) and a CPSS. The researchers were systematically assessed a number of effectual parameters of energy and exergy performance, economic analysis, and water productivity while considering four water depths. The results showed the consistent performance of MPSS unit compared to the CPSS unit. In this regard, the MPSS attained an optimum water productivity of 4.10 L/m^2 day at a water depth of 2 cm, in a comparison to 3.05 L/m^2 day for the CPSS. The water productivity of integrated MPSS and PHP was increased by 34.42%, 37.13%, 31.87%, and 29.62% for water depths of 2, 3, 4, and 5 cm, respectively. Also, other enhancements of 25.51% and 29.31% were deduced for energy and exergy efficiencies for the MPSS if compared to the CPSS at 2 cm depth.

The daily water productivity of a stepped pyramid solar still (SPSS) was enhanced by Mohammed et al. (2024) who integrated it with a solar parabolic dish (SPD). The suggested system includes a multi-phase solar still (MUPSS), a CPSS, a photovoltaic panel, an SPD with a spiral coil, an Arduino Mega controller, a circulating pump, and an SPSS. The results signified a daily water productivity of 6,038 mL for the MUPSS, that is 131% higher than the daily water productivity of CPSS (2,612 mL). The MUPSS attained a peak thermal efficiency of 56%, in a comparison to only 24% for the CPSS. Furthermore, the MUPSS ascertained an improvement of exergy efficiency of 4.26% against 3.79% for the CPSS. The estimated water production cost for the MUPSS was just $0.01635.

The optimisation of operating parameters a PSS technology was conducted by Godwin et al. (2024) using the Response Surface Methodology (RSM) and feed forward Artificial Neural Networks (ANN). The researchers intended to maximise the thermal performance and water productivity. The researchers optimised the inclination angle, solar intensity, and water depth, which resulted in a thermal efficiency improvement of 42% and an increase of water productivity of 1.8 L/m^2. The freshwater production cost was lessened to 0.20 INR due to attaining these improvements.

A comparison between two solar still designs of a CPSS and an MPSS with a PHP was conducted by Pandey and Naresh (2025) who demonstrated experimental and analytical research. Specifically, the basin water is heated externally via a PHP and solar concentrator (SC) in the MPSS.

The models of Zheng Hongfei, Kumar and Tiwari, Chen, and Dunkel were utilised to predict the water productivity that deduced cumulative water productivity errors of 5.81% for MPSS and 5.9% for CPSS. The average basin water temperature in the MPSS reached 64.56 °C, compared to 58.8 °C in the CPSS, attributed to additional heating from the PHP.

Afshari et al. (2025) proposed a novel seawater heating method using a spiral-type solar heater to raise saltwater temperatures. The approach included cotton-based materials, a perforated delivery pipe, and intermittent pumping to maximise heat absorption and evaporation. This setup facilitated more efficient condensation in a triangular prism-shaped condensation unit. The results showed a 138.56% increase in distilled water output and a 128.57% improvement in thermal efficiency. Exergy efficiency improved by 152.63% compared to the reference setup. Table 2 summarises the relevant studies concerning external modifications in triangular solar stills (TSS).

Internal and external modifications

SSs incorporating different basin heat storage materials were evaluated for productivity enhancement by Riahi et al. (2015). Three identically shaped triangular solar stills were fabricated for the study. One still used a stainless steel trough as the basin, while another employed a clear polyethylene film as a cover. Each basin measured 50 cm in length, 30 cm in width, and 8 cm in depth. The configurations differed in terms of basin treatment, one lined with black paint, another with a 2 cm layer of black soil, and one left untreated as a control.

Experimental results obtained under tropical Malaysian conditions indicated that basin heat storage materials significantly improved freshwater production. The solar still with the black-painted basin achieved the highest yield, producing 101% more water than the standard still, and 20% more than the still lined with black soil, according to the daily cumulative water output data. Figure 15 illustrates the variation in total water production across the three stills on a representative day, July 14, 2014. During this day, SS1 (WC1) yielded the lowest amount of water.

Two types of SSs, a triangular design and a tubular design—were experimentally evaluated by Rahbar et al. (2018) under real-world weather conditions (Fig. 16). The researchers inspected the effects of ambient temperature and solar radiation on water productivity and the system efficiency. Also, the computational fluid dynamics (CFD) was applied to numerically model the fluid flow, detecting the complicated air circulation patterns within the enclosures. The results stated that the tubular still can outpace the triangular one by around 20%. Numerical simulations indicated that the tubular still achieved superior water output due to lower entropy generation and more pronounced recirculating zones. However, the triangular solar still was found to produce freshwater at a lower cost, primarily due to its simpler manufacturing process. Based on the experimental data, two novel empirical correlations were proposed to predict water productivity in both configurations. As illustrated in Fig. 17, the maximum overall efficiency was 41% for the tubular still and 35% for the triangular still.

A MPSS was constructed by Fallahzadeh et al. (2020) and experimentally tested in Mashhad, Iran (latitude 36°18', longitude 59°34'), using a CPSS integrated with a Heat Pipe Solar Collector (HPSC). Three distinct filling ratios (FRs) were tested using two working fluids, ethanol and water. The use of the HPSC increased the temperature gradient between the glass cover and the basin water. The cumulative yields of the MPSS (with water as the working fluid at a 40% FR) and the CPSS were enhanced to 6970 ml/m^2 and 3300 ml/m^2, respectively. Although the immediate efficiency remained relatively constant, the inclusion of the HPSC significantly increased the exergy efficiency. Among the tested conditions, the best MPSS performance was achieved using water at a 40% FR, though ethanol at the same FR also delivered commendable results. The cost–benefit analysis further confirmed the viability of the MPSS, with the system delivering freshwater at a cost of only $0.0137 per liter.

Kabeel and Abdelgaied (2020) enhanced the performance of PSSs by employing an absorber plate made from graphite, which has high thermal conductivity, and by cooling the glass cover. A 25 mm thick graphite absorber was utilised as a sensible heat storage medium, enable of absorbing solar radiation throughout peak hours and releasing it during the evening. The glass cover was superficially cooled to accelerate the vapor condensation and therefore enhance the water productivity. Specifically, the experiments were achieved while considering the efficiency of a CPSS and an MPSS, one with the high-TC absorber and another with a cooled glass cover. The daily fresh water productivity of the MPSS was estimated to be ranged between 9 to 9.19 L/m^2 under similar environmental conditions, if compared to the daily water production of CPSS of 4.37 to 4.43 L/m^2. Statically, this is an improvement of 105.9% to 107.7% for the MPSS layout if compared to the CPSS.

Maatki (2022) presented a 3D numerical research on doubly diffusive convection of the air-vapor mixture within a TSS. The researchers evaluated the impacts of adding Nanofluids for a modified layout of the seawater feed system. The effectual parameters of Rayleigh number, buoyancy ratio, Nanofluid volume fraction (0–10%), and secondary seawater feed length (Lp = 0.1–0.5) were tested throughout the experiments. With an Lp of 0.5 and a height

Table 2 A summary of relevant researches on TSS with external modifications

Location	Modification	Key performance metrics	Results and remarks	Authors (year) [reference]
Tamil Nadu, India	Triangular pyramid solar still integrated with inclined solar still with baffles	Nighttime yield increase: 46% to 86%	The yield of triangular pyramid solar still is increased by integrating it to an inclined solar still with baffles	Kumar et al. (2016)
Tamilnadu, India	PSS with biomass heat source	Productivity increase: 48% (sensible heat storage), 19% (evaporative materials)	The substantial, sensible heat storage materials produce 48% more productivity than conventional still. Also, evaporative materials produce 19% more productivity than conventional still	Rajan et al. (2016)
Tamil Nadu, India	PSS integrated with inclined solar still with baffles	Daytime yield decrease: 6% to 46%; nighttime yield increase: 46% to 86%	The increase in water mass decreases the yield from 6 to 46% during the daytime, whereas due to thermal energy storage at higher mass, the yield during the nighttime increases from 46 to 86% with integration	Kumar et al. (2016)
Tamil Nadu, India	PSS with forced cover cooling techniques	Yield improvement: 104% (wind speed 0.5 to 4 m/s)	The yield of fresh water depends on naturally occurring wind velocity and the results show that the improvement in yield is 104% with an increase in velocity from 0.5 to 4 m/s	Nagarajan et al. (2016)
Tamil Nadu, India	Integrating inclined solar still with triangular pyramid solar still	Yield improvement: 79.05%	Higher water temperature is achieved in the case of inclined solar still integrated with triangular pyramid solar still at the minimum water depth of 0.02 m	Kumar et al. (2017)
Tuticorin, India	Stepped basin PSS integrated with solar pond	Productivity increase: 66.2%	The productivity of the modified solar still was 4,920 mL per day, which was 66.2% higher than the conventional solar still	Appadurai and Velmurugan (2017)
Al Ain City, United Arab Emirates	Passive PSS with an automatic feed water system	Maintained optimal water level	The proposed feedwater system can maintain the water level inside the solar still to its optimum value that improves the solar still productivity and convenience for the end-user	Elgendi et al. (2022)
Gazipur, Bangladesh	Triangular shape solar distillation system with solar PV panel	Higher production rate	The production rate of the still is higher than the conventional one	Ahsan et al. (2023)
Chengdu, China	Pyramid floating solar still with enhanced condensation surfaces	Yield increase: 16.76%; CPL: 0.129 $/L	The highest yield of nano-coated FSS was 1.017 kg/m2/d under an accumulated irradiation of 15.166 MJ/m2/d, which was 16.76% higher than that of uncoated FSS	Wei et al. (2024)
Surat City, India	PSS with pulsating heat pipe	Yield improvement: 34.42%–29.62%; energy/exergy efficiency increase: 25.51%/29.31%	Improvement in the yield of MPSS due to the attachment of PHP was 34.42%, 37.13%, 31.87%, and 29.62% as compared to CPSS for 2 cm, 3 cm, 4 cm, and 5 cm, respectively	Pandey and Naresh (2024)
Faculty Engineering, South Valley University	Solar parabolic dish integrated with stepped pyramid solar still	Daily yield increase: 131%; peak energy efficiency: 56%	The daily productivity of freshwater from MPSS was greater than that from CPSS and reached 6038 and 2612 mL/day, respectively, with an increase in the daily yield of 131%	Mohammed et al. (2024)
Vellore, India	PSS with response surface methodology	Thermal efficiency increase: 42%; water yield improvement: 1.8 L/m2	The thermal efficiency increased by 42%, while the water yield improved by 1.8 L/m2	Godwin et al. (2024)
Surat, India	PSS integrated with pulsating heat pipe	Basin water temperature: 64.56 °C (MPSS) vs. 58.8 °C (CPSS)	The theoretical average basin water temperature observed in MPSS is 64.56 °C, and CPSS is 58.8 °C	Pandey and Naresh (2025)
Erzurum, Turkey	Triangular prism solar still with spiral heating	Distilled water increase: 138.46%; efficiency improvement: 128.57%	Comparing the optimal experiment with the reference experiment revealed a substantial 152.63% increase in exergy efficiency	Afshari et al. (2025)

of 0.2, thermal performance augmented by 38%, and mass transfer performance at the condensing and evaporating surfaces enhanced by 55%. The heat transfer improvement was affected by how the solar stills was operated and the Nanofluid concentration used.

Emran et al. (2022) developed TSSs to improve the water productivity in the Malaysian climate. The TSSs contained an external PVC pipe solar heater and an internal detached condenser. The results ascertained an increase of water productivity of 24% in a comparison to passive systems.

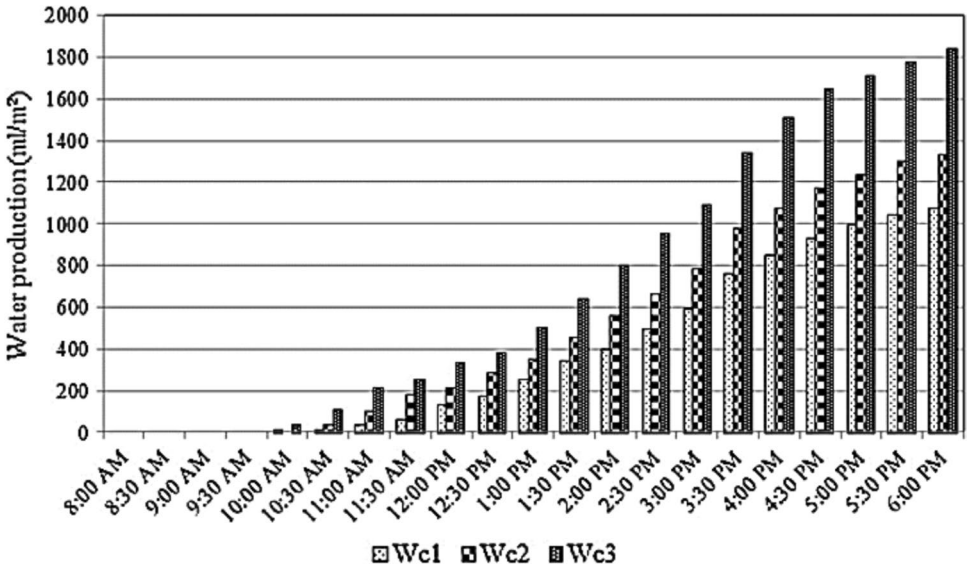


Fig. 15 Variations in three solar stills' total production of water from (8 am) to (6 pm) upon a normal day on July 14, 2014 (Riahi et al. 2015)

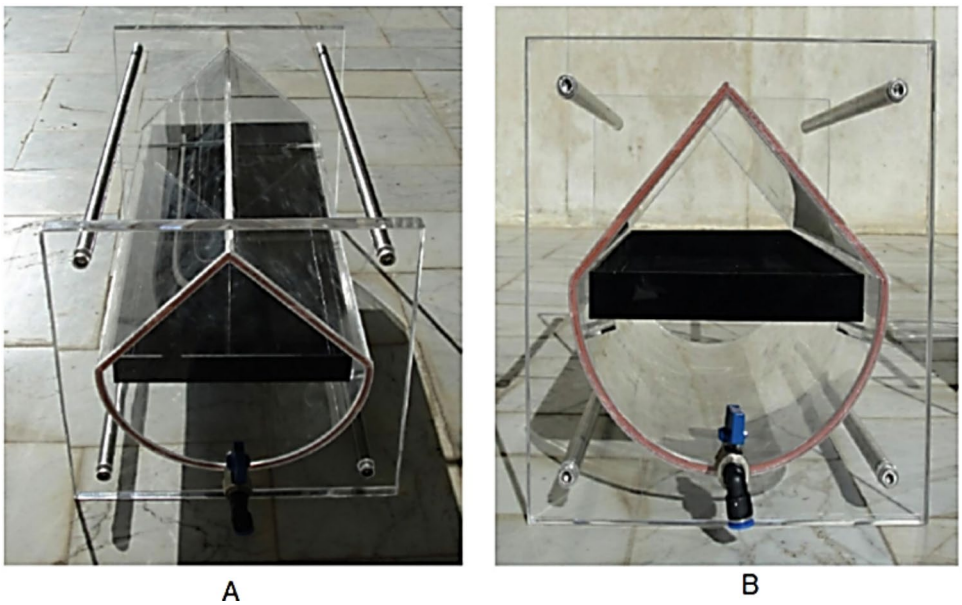


Fig. 16 Triangular solar still in various perspectives (Rahbar et al. 2018)

However, it was ascertained that the water productivity can be impacted by ambient air and water temperatures, solar radiation intensity, and relative humidity. In this aspect, robust relationships were deduced that correlate the daily water productivity and average water temperature, average ambient temperature, and solar radiation. The predicted freshwater production cost per liter was economically viable.

Sharshir et al. (2022) improved PSS performance using external condensers, nanoparticles, evacuated tubes, and ultrasonic foggers. The experimental setup compared MPSSs integrated with these features against CPSSs. The MPSS with five evacuated tubes and an external condenser increased freshwater output, exergy efficiency, and energy efficiency by 91.09%, 18.48%, and 45.26%, respectively. Integrating 1 wt.% of Carbon Black (CB) nanoparticles can boost the water productivity by 132.86%, 28.22%, and 75.43%. The inclusion of three Nanoparticle-improved ultrasonic foggers raised the enhancements to 162.15%, 34.26%, and 81.51%, respectively. Interestingly, these improvements achieved a reduction of fresh water production cost by more than 32.04% in a comparison to the CPSS.

Hammoodi et al. (2023) investigated the effects of various wick materials on the overall performance of PSS technology. The researchers built and experienced two pyramid solar stills, a CPSS and a Wick-type Modified Pyramid Solar Still (WPSS), while considering similar conditions. Wick materials tested included thick cotton, light cotton, velvet, and jute fabrics. The findings indicated that Light Cotton Fabric (LCF) can attain the greatest water productivity

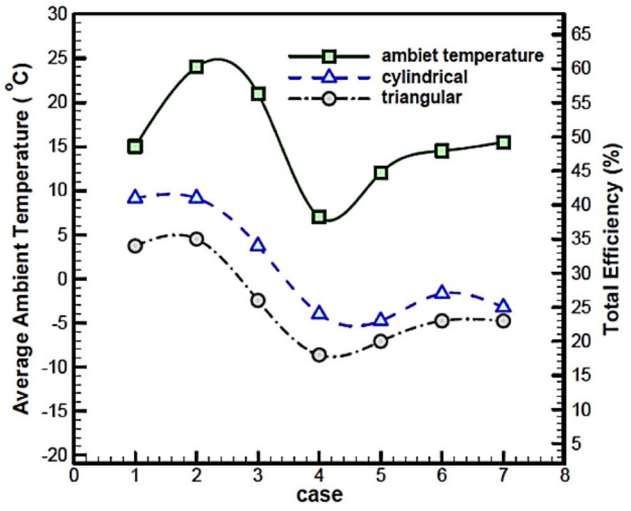


Fig. 17 Average ambient temperature's impact on overall efficiency (Rahbar et al. 2018)

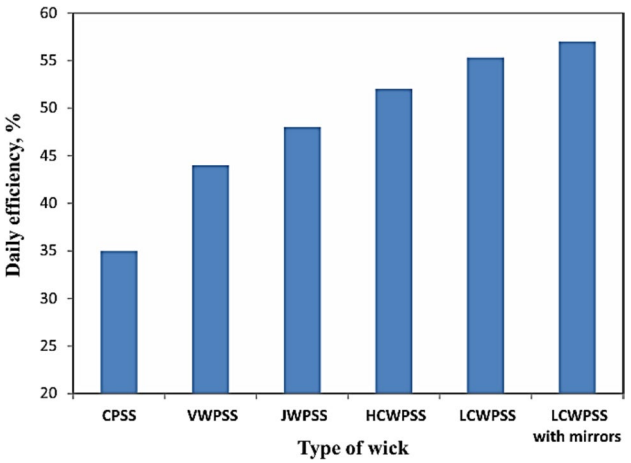


Fig. 18 Daily water productivity for different wick materials (Hammoodi et al. 2023)

and thermal competence for the WPSS. The results ascertained a peak efficacy of 55.3% and a water productivity of 1,476 mL/m^2 day. More importantly, the results indicated the increase of water productivity with the use of LCF by 136.6% in a comparison to CPSS. This was the greatest improvement made for the whole tested wick materials. The CPSS preserved a fixed efficacy of 34.5%, while WPSS effectiveness was changed based on wick type and operational conditions, as seen in Fig. 18.

Using a 4E (exergy, energy, environmental, and economic) analysis, Alharbi and Alshamrani (2024) enhanced the thermal efficiency of the Pyramid Solar Distiller (PSD). The Modified Pyramid Solar Distiller (MPSD) was evaluated both with and without heaters, and with varying water volumes ranging from 6.2 to 49.6 L in the basin. Trays were suspended on three vertical walls, and the system was tested using an external condenser and a variable-speed fan (0.25–1.5 rpm). The results showed that the MPSD increased yield by 33% and thermal efficiency by 50.2% when operated with a 1 cm water depth and a 12.4 L basin volume. Furthermore, while the total productivity of the PSD was 2820 mL/m^2, the MPSD achieved 3755 mL/m^2, and the MPSD with heaters reached 6430 mL/m^2—an increase of 98.5%. The optimal performance was attained at 1.25 rpm, yielding a 133% increase in productivity, a 66.5% rise in thermal efficiency, and a 4.58% improvement in exergy efficiency. The cost of water production for the MPSD was $0.23/L, compared to $0.34/L for the PSD. Depending on the configuration, CO_2 emissions ranged from 30.93 to 32.23 tons per year, while the enviro-economic indicators ranged from 447.46 to 467.33 tons per year.

Esfe and Toghraie (2024) explored the impact of glass surface and saline water temperatures on water productivity in Kerman Province (Fig. 19). The researchers considered a number of effectual parameters like mass fraction, temperature, streamlines, velocities, average Nusselt number, and water productivity under various conditions. The

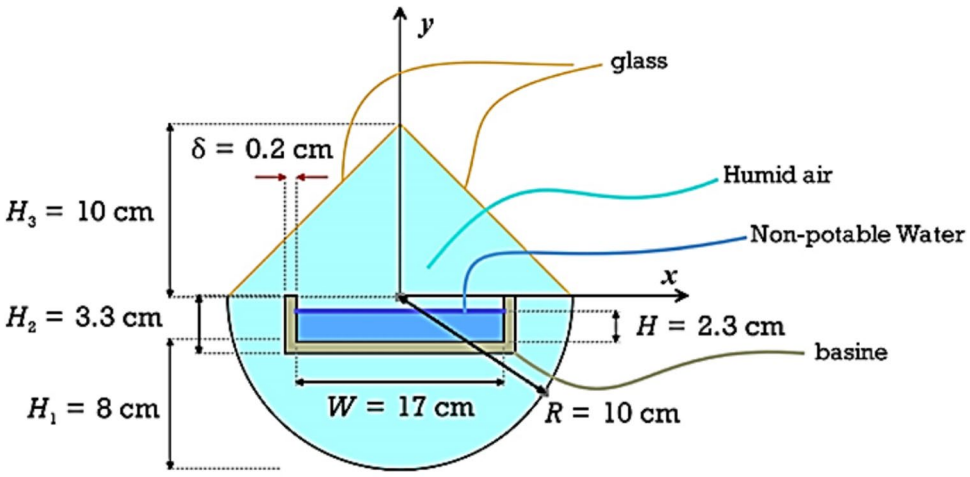


Fig. 19 Setup in geometry (Esfe and Toghraie 2024)

results stated that an increase in saline water and glass surface temperatures can enhance the water productivity. This clear enhancement was ascribed to an improved density gradient. The greatest water productivity was registered when the water temperature was 65 °C and glass temperature was 61 °C.

The impact of square baffles on the overall performance of a MSS design was inspected by Elamy et al. (2024b). To successfully improve the evaporation and condensation rates, the integration of reflectors was studied. Also, the efficacy of a PCM composed of silver Nanoparticles and paraffin wax was examined within the advanced solar still. The economic viability of the suggested system was evaluated based on a thermo-economic evaluation. The findings ascertained a notable rise in water productivity. The MSS with square baffles attained a water productivity of 9800 mL/m^2 day. This is specifically a higher water productivity of 193% compared to the water productivity of 3550 mL/m^2 day of the reference solar still. Furthermore, the advance solar still of integrated square reflectors and Nano-PCM can achieve an increase of water productivity of 265% while attaining the optimum layout of 25 wick cords. This layout also achieved an efficacy of 63%. Referring to the economic analysis, it was assured a water production cost for the modified SS with Nano-PCM and square baffles of only $0.01/L in a comparison to $0.014/L for the reference solar still. Environmentally, the modified system also resulted in a reduced annual CO_2 emission of 28.8 tons. Figure 20 shows that the CPSS equipped with 25 wick cords can attain a remarkable efficiency of 49.5%, demonstrating a 122% enhancement over the PSS without baffles.

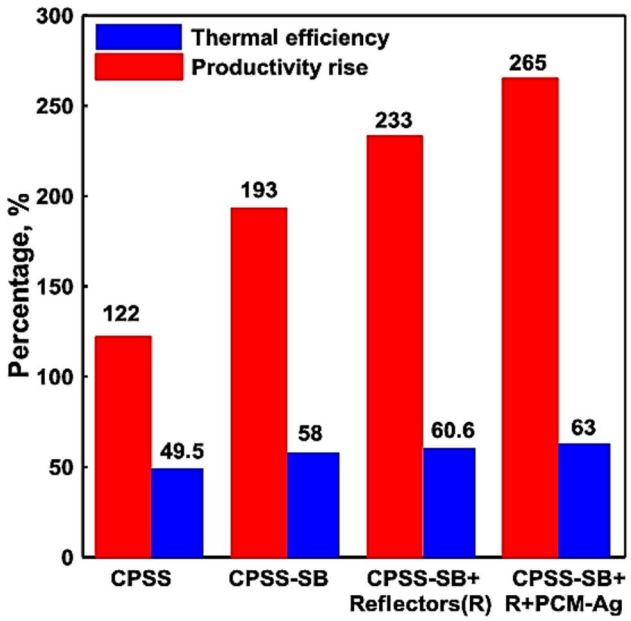


Fig. 20 The rise in the manufacture of cord PSS and effectiveness beneath various conditions (Elamy et al. 2024b)

Alshamrani (2024) presented a comprehensive study on performance enhancements of Pyramid Distiller Solar Stills (PDSS) through both internal and external modifications. The modified configuration, termed Modified Pyramid Distiller Solar Still (MPDSS), incorporated suspended V-shaped internal steps to expand the evaporative surface area. Additional modifications included the application of external reflectors (MPDSS-R), a Nano-enhanced thermal storing material (MPDSS-NTSM) composed of paraffin wax doped with CuO nanoparticles, a Peltier cooling device (MPDSS-P), and a forced external condensation system with a fan (MPDSS-C). The experimental comparison with a conventional PDSS demonstrated notable productivity improvements: 40% with suspended V-steps, 96% with reflectors, 139% with the thermal storage material, 143% with external condensation, and 148% with Peltier-assisted cooling. Among all configurations, MPDSS-P achieved the highest thermal efficiency (54.67%) and exergy efficiency (4.87%). Furthermore, the cost of freshwater production was significantly reduced from $0.343/L for the baseline PDSS to $0.082/L for the most enhanced system. The researchers also evaluated environmental impacts, revealing that advanced configurations decreased annual CO_2 emissions and yielded better enviro-economic indicators. Alshamrani concluded that a synergistic combination of geometric design improvements, advanced materials, and active thermal and condensation strategies can greatly enhance the feasibility and sustainability of solar desalination technologies. Future research directions include optimizing step geometries, investigating the longevity of thermal storage materials, and exploring alternative passive cooling techniques. Figure 21 shows the hourly and total distillates for MPDSS and PDSS using Peltier and reflectors.

In an effort to address the supply of potable water, Fadhil and Abed (2024) investigated the use of solar energy. A pyramid-shaped solar water purifier was employed in Anbar City, Iraq, featuring a 30° tilted glass cover and a basin area of 1 m^2. During the 24-h testing period, the temperatures of the surrounding air, saline water, and glass cover peaked at midday. The results indicated that basin water temperature significantly influenced distillate water production, with output increasing in response to rising water temperatures. Notably, there was a 60% reduction in water output when the water depth in the solar still increased from 2 to 10 cm.

Essa et al. (2025) enhanced the performance of a PSS by incorporating a vertically positioned triangular absorber and a single-axis solar tracking system. The modified pyramid distiller (MPD) with a triangular-shaped absorber (TSA) exhibited a 32% increase in productivity; with a corrugated TSA, productivity rose by 61%, and with solar tracking, it increased by 102%. Compared to the standard PSS, the MPD equipped with reflectors and a corrugated

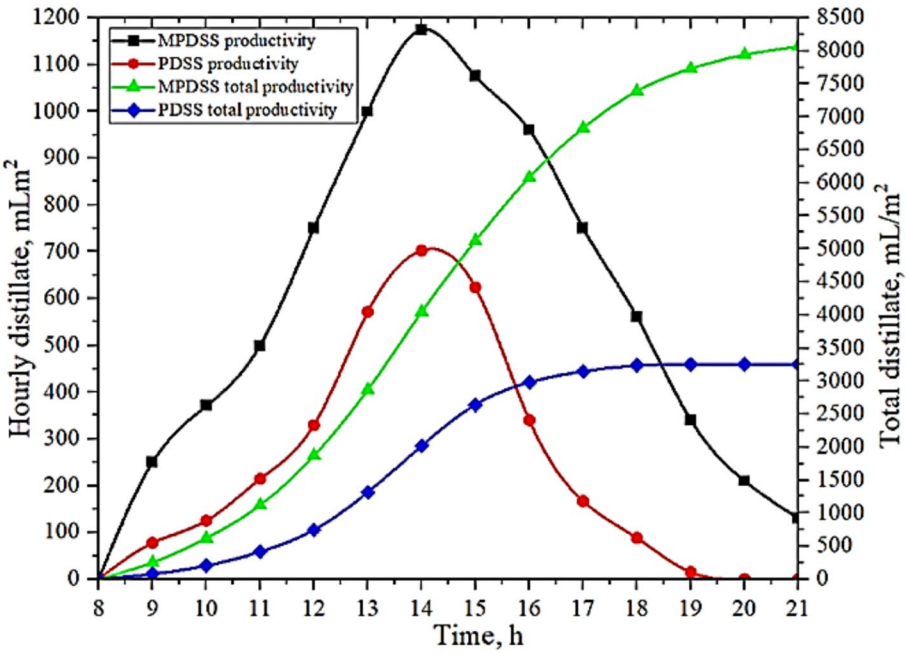


Fig. 21 Hourly and total distillates for MPDSS and PDSS using Peltier and reflectors (Alshamrani 2024)

TSA achieved a maximum distillate output of 10,400 mL/m^2/day, particularly with the addition of PCM. The energy efficiency of the following modified SS configurations was also evaluated: the MPD with a triangular absorber (39.6%), the MPD with a corrugated triangular absorber (49%), the MPD with a reflector and absorber (50.5%), the MPD with absorber, fan, reflector, and PCM (59%), and the MPD with reflector, absorber, and power conversion module (57%). Table 3 presents a summary of the relevant research on TSS incorporating both internal and external modifications.

Comparative analysis of PSS technologies

Table 4 presents a comparison of PSS technologies that incorporate internal modifications, external modifications, and combined modifications simultaneously. The comparison evaluates key design changes and performance indicators, including productivity improvements and efficiency gains, along with their associated advantages and limitations. These design strategies have demonstrated effectiveness in addressing freshwater scarcity by enabling the development of more efficient solar desalination systems.

Table 5 demonstrates an intuitive impression of the techno-economic and environmental performance of different layouts of PSS. Clearly, the water production cost indicates a notable change across the tested layouts. For example, the integrated system of PSS with heat pipe solar collector delivers the lowest fresh water production cost of $0.0137, that is considerably lesser than the traditional layout like the conventional solar still of $0.34. Furthermore, the modified solar still (MSS) introduced a notable decrease of 20.05 tons/year in CO_2 emissions in a comparison to the CSS's 13.668 tons/year. Indeed, the combination of Nano-materials and optimum layouts like those using nanofluids and PCM, would aids to ensure improved thermal performance at reduced payback periods. Overall, these improvements would underscore the potential for enhanced sustainability and cost-effectiveness in solar desalination technologies.

Critical analysis of PSS with internal and external modifications

The global water shortage has strengthened the request for sustainable and effective water desalination methods. In this regard, the PSS technology stands out as a hopeful solution as a result to their competency to capture the solar energy to assure water productivity. Upgraded enhancements in PSS units were intended to test the feasibility of internal and external enhancements to elevate the overall performance, water productivity, and cost-effectiveness. This section delivers a critical analysis of these enhancements by assessing their influence on the effectual performance indicators, scalability, and practical applicability. This review emphasized the limitations of modified PSS systems while outlining the future development opportunities. However, it should be stated that the incorporation of various improvements in PSS can lead to improved water productivity via employing different consistent techniques. These techniques were positively affect the thermal management at improved surface area, and included phase change dynamics. Explicitly, the incorporation of PCMs into PSS

Table 3 A Summary of relevant studies on TSS with internal and external modifications

Location	Modification	Key performance metrics	Results and remarks	Authors (year) [reference]
Perak, Malaysia	TSS with transparent polythene film and stainless steel trough cover/basin	Daily cumulative water production, productivity percentage increase	Solar still with black-painted basin achieved 101% higher productivity than conventional still and 20% higher than solar still with black soil in basin	Riahi et al. (2015)
Semnan, Iran	Triangular vs. tubular solar stills	Water productivity, total efficiency, entropy generation, manufacturing cost	Tubular still outperformed triangular still by 20% in productivity. Total efficiency: 41% (tubular) vs. 35% (triangular). Lower cost for triangular still	Rahbar et al. (2018)
Mashhad, Iran	PSS integrated with heat pipe solar collector (HPSC)	Accumulated yield, exergy efficiency, economic cost	MPSS with water (FR = 40%) produced 6970 mL/m2 vs. CPSS (3300 mL/m^2). Cost: $0.0137/L for MPSS	Fallahzadeh et al. (2020)
Tanta University, Egypt	PSS with graphite absorber plate and glass cooling	Daily water production, thermal efficiency	Modified still achieved 9–9.19 L/m2/day vs. 4.37–4.43 L/m2/day for traditional still (105.9–107.7% improvement)	Kabeel and Abdelgaied (2020)
Monastir, Tunisia	TSS with Nanofluid and seawater feed design	Thermal and mass performance enhancement	Second-stage seawater supply improved thermal performance by 38% and mass performance by 55%	Maatki (2022)
Selangor, Malaysia	TSS with PVC pipe solar heater and internal condenser	Daily water production, desalinated water cost	Active still produced 24% more water than passive still. Cost per liter: $0.03 (reasonable)	Emran et al. (2022)
Kafrelsheikh, Egypt	PSS with evacuated tubes, condenser, nanoparticles, and foggers	Freshwater output, energy/exergy efficiency, cost reduction	MPSS with 1 wt.% CB nanoparticles achieved 132.86% higher output. Cost reduced by 32.04%	Sharshir et al. (2022)
Karbala, Iraq	PSS with wick materials and reflectors	Productivity, thermal efficiency	Light cotton wick improved productivity by 136.6% vs. CPSS. Thermal efficiency: 55.3% (light cotton) vs. 34.5% (CPSS)	Hammoodi et al. (2023)
Ha'il City, Saudi Arabia	PSS with trays, heaters, and forced condensation	Productivity, thermal/exergy efficiency, CO2 emissions	Best performance at 1.25 rpm: 133% productivity increase, 66.5% thermal efficiency, 4.58% exergy efficiency. Cost: 0.23/L(MPSD)vs.0.23/L(MPSD)vs.0.34/L(PSD)	Alharbi and Alshamrani (2024)
Kerman province	Triangular double-slope solar still	Water production rate, temperature gradient	Higher water–glass temperature difference increased freshwater production. Max rate at Tw = 65 °C, Tg = 61 °C	Esfe and Toghraie (2024)
Arar, Saudi Arabia	PSS with square baffles, reflectors, Nano-PCM	Distillate yield, efficiency, CO2 emissions	Modified still achieved 9800 mL/m2 day vs. 3550 mL/m2/day (193% increase). Efficiency: 63%. Annual CO_2: 28.8 tons	Elamy et al. (2024b)
Taif, Saudi Arabia	PSS with suspended v-steps, reflectors, Peltier	Productivity, thermal efficiency	MPDSS-P (Peltier) produced 8070 mL/m2 day vs. PDSS (3250 mL/m2 day). Thermal efficiency: 53.96%	Alshamrani (2024)
Anbar city, Iraq	PSS under Iraqi climate	Water depth impact on productivity	Water production decreased by 60% when depth increased from 2 to 10 cm	Fadhil and Abed (2024)
Kafrelsheikh, Egypt	PSS with tracking, corrugated absorber, PCM	Productivity, energy efficiency	Highest productivity: 10,400 mL/m2 day (166% increase). Energy efficiency: 57–59% with PCM and reflectors	Essa et al. (2025)

has deduced an effectual absorption of excess heat during peak solar radiation periods besides releasing it during off-peak times. Overall, this approach has ensured a stable temperature within the PSS, maintaining optimal operating conditions of upgraded evaporation rate while prevention heat loss. Furthermore, the inclusion of Nanoparticles into PSS has enhanced the thermal conductivity of the water. In turn, this would permit for faster heat transfer from the absorber to the water, and therefore quickening the evaporation process. Moreover, the utilisation of advanced absorber materials of high surface area would raise the efficient area for solar absorption. Indeed, this would increase the overall heat absorption besides enhancing the evaporation surface area, and consequently the water production rate. The integration of reflectors and collectors into PSS to produce hybrid systems would deliver additional sunlight onto the absorber, causing a greater temperature at improved evaporation rates.

Referring to the internal modifications of PSS, which primarily targeted the evaporation and condensation phases, the key innovations and their limitations are as follows:

Table 4 Comparative analysis of PSS technologies

Technology Type	Key Modifications	Performance Metrics	Advantages	Limitations
Internal Modifications				
PCM Integration	Paraffin wax, quartz rock, or Nano-enhanced PCM under absorber plate (Kabeel et al. 2020, Chauhan et al. 2023)	– 101.5% productivity increase with PCM (Kabeel et al. 2020)—64.25% yield boost with PCM + clay pots (Chauhan et al. 2023)	Extends operational hours by storing thermal energy, improves nocturnal yield	Adds weight and complexity; potential leakage risks
Nanoparticles	TiO_2, CNT-water Nanofluids, or carbon black (Kabeel et al. 2019, Ghachem et al. 2021)	– 6.1–54.4% efficiency improvement—Enhanced heat transfer with CNT-water Nanofluids	Improves thermal conductivity and evaporation rates	Nanoparticle stability and long-term durability concerns
Fins and Wick Materials	Hollow copper fins, wool cloth wicks, or jute/cotton cords (Modi and Shah 2022, Alawee et al. 2021)	– 29.34% efficiency gain with submerged wick-fins—76.7% productivity increase with light cotton wicks	Maximizes evaporation surface area; cost-effective	Material degradation over time; maintenance required
External Modifications				
Solar Collectors and Reflectors	Integrated solar parabolic dishes or mirrors (Mohammed et al. 2024, Abdullah et al. 2020)	– 131% productivity increase with solar dish (Mohammed et al. 2024)—62.5% boost with reflectors (Asadabadi and Sheikholeslami 2022)	Enhances solar radiation absorption; scalable for large systems	Higher initial cost; dependency on sunlight intensity
Forced Condensation	External condensers with fans or Peltier devices (Essa et al. 2024, Alshamrani 2024)	– 194% yield increase at 2 rpm fan speed—148% improvement with Peltier cooling	Accelerates vapor condensation; improves efficiency	Energy-intensive; requires auxiliary power sources
Floating Designs	Nano-coated floating stills (Wei et al. 2024)	– 16.76% higher yield than uncoated systems	Reduces land dependency; efficient for coastal regions	Susceptible to environmental wear; higher fabrication costs
Combined Modifications				
PCM + Reflectors	Hybrid systems with PCM and external reflectors (Abdullah et al. 2020, Essa et al. 2025)	– 166% productivity increase (Essa et al. 2025)	Synergistic thermal storage and light concentration; high efficiency	Complex integration; requires precise alignment
Nano-PCM + Baffles	Nano-enhanced PCM with square baffles and reflectors (Elsmy et al. 2024)	– 193% yield improvement—63% thermal efficiency	Optimizes heat distribution and condensation	High material and maintenance costs
Rotational Systems	Rotating absorber plates (Baiee et al. 2024)	– 30.2% productivity increase at 1 rpm	Enhances turbulence and heat transfer	Mechanical complexity; energy consumption for rotation

- PCMs, such as paraffin wax, absorb solar energy during peak radiation and release stored heat during off-sunlight hours, extending operational time. Studies report productivity gains ranging from 35% to 101.5%. However, high cost and challenges in ensuring uniform heat distribution.
- Nanoparticles such as TiO_2 and carbon nanotubes can enhance the thermal conductivity and solar absorption. TiO_2-coated absorbers presented an increase in water productivity that ranges between 6.1% and 54.4%.
- Integration of quartz rocks and mild steel scraps can improve the thermal storage that stated an increase in water productivity of around 30%. However, this practice requires a considerable maintenance.
- The utilisation of wick materials such as light cotton and jute and aluminum hollow fins has improved the evaporation rate and overall performance. In this aspect, the water productivity was increased to 136.6% while using light cotton wicks in a comparison to standard solar stills systems. However, possible fouling on the fins can be deduced via the use of organic wick materials.

Regarding the implication of external enhancements of PSS, that principally concentrated on upgrading solar energy collection and condensation effectiveness, the main revolutions and their challenges are stated below:

- The water productivity can be increased by more than 194% by the inclusion of parabolic dishes or inclined solar collectors with boosting the thermal efficiency to 62.4%. However, the reliance on favorable solar angles with a high capital cost are stated as the main challenges.
- External condensers with fans can improve the water productivity by more than 133%. However, further power requirements can constrain their implications in off-grid or remote settings.

Table 5 A summary of techno-economic and environmental data for PSS

Modification type	configuration / study	Cost per Liter (CPL)	Payback period	CO_2 Reduction / Environmental note
Internal	PSSCAP (Corrugated Absorber) (Ghandourah et al. 2022)	0.53 INR	–	–
	PSS with Fins & Insulation (Asadabadi and Sheikholeslami 2022)	–	–	Cost reduction of 12.76%—7.33%
	PSS with Nano-PCM, Black Sand, Sponge (Case III) (Atiua Anika et al. 2024)	–	126 days	–
	Conventional Solar Still (CSS) (Atiua Anika et al. 2024)	–	185 days	–
External	PSS with Pulsating Heat Pipe (MPSS) (Nagendra Pandey 2024)	–	–	–
	Floating PSS (Nano-coated) (Wei et al. 2024)	$0.129	–	–
	Stepped PSS with Solar Parabolic Dish (MUPSS) (Mohammed et al. 2024)	$0.01635	–	–
	PSS with Response Surface Methodology (Godwin et al. 2024)	Reduction of 0.20 INR	–	–
Combined (Internal and External)	PSS with Heat Pipe Solar Collector (HPSC) (Fallahzadeh et al. 2020)	$0.0137	–	–
	Modified PSD (MPSD) (Essa et al. 2024)	$0.18	–	–
	Modified PSD (MPSD) (Alharbi and Alshamrani 2024)	$0.23	–	30.93–32.23 tons/year
	Conventional PSD (Alharbi and Alshamrani 2024)	$0.34	–	–
	PSS with Square Baffles & Nano-PCM (Elamy et al. 2024b)	$0.01	–	28.8 tons/year (annual emissions)
	Conventional PSS (Reference) (Elamy et al. 2024b)	$0.014	–	–
	Modified PDSS (MPDSS-P) (Alshamrani 2024)	$0.082	–	Reduced annual emissions
	Conventional PDSS (Alshamrani 2024)	$0.343	–	–
	Modified Solar Still (MSS) (Mohamed Ragab Diab 2025)	45.45% lower than CSS	7.2 years	20.05 tons/year (mitigated)
	Conventional Solar Still (CSS) (Mohamed Ragab Diab 2025)	–	17.18 years	13.668 tons/year (mitigated)
	Hybrid PVT PSS (Mohtasim et al. 2025)	~94.13% reduction	–	6.98 tons (over operational lifetime)

- Integrating PCMs with reflectors or solar collectors can enhance the water productivity to more than 166%. However, the system complexity and upfront investment are the most challenges.
- An increase of 16.76% of water productivity was assured via the use of floating PSS units with Nano-coated surfaces, besides improving the overall efficiency by 30.2% due to using rotational mechanisms. However, the concerns of durability as a result to contact to harsh environmental conditions can be the key challenge.

Moreover, several qualified visions can be stated as follows;

- The integration of internal and external improvements such as PCMs, reflectors, and condensers can deduce the greatest water productivity of more than 166%.
- The reflective coatings can reduce the overall water production cost and associated complexity and therefore can ascertain an advanced thermal storage solution.
- The inclusion of nanoparticle coatings can provide a notable solution for prevalent adoption in a comparison to parabolic collector systems.

Conclusions

A detailed examination of pyramid (triangular) solar stills with internal and external modifications was the main focus of the current study, that established their promising aspects as sustainable solutions for elevated water productivity. The revised systems utilised solar energy to resolve the global water shortage as a cost-effective and environmentally friendly process. This review stated that the integration of internal improvements like the inclusion of PCMs, nanoparticles (e.g., TiO_2 and CNT-water Nanofluids), and energy storage materials (e.g., paraffin wax and quartz rock), can meaningfully enhance the water productivity. Statistically, the incorporation of PCMs has augmented the water desalination performance by more than 101.5%. Also, the use of nanoparticle can enhance the effectiveness by 6.1% to 54.4%. Also, the addition of external modifications such as the installation of solar collectors, reflectors, and forced condensation systems can successfully enhance the overall efficiency. Statistically, the combination of reflectors and PCMs can lead to an improvement of 194% in water productivity while attaining 62.4% of thermal efficiency. Moreover, it was stated that the use of hybrid systems that combine multiple enhancements like PCMs with reflectors

or Nanofluids with external condensers can assure the highest performance results. These practices resulted in an efficiency improvement of more than 166%, highlighting the synergistic advantages of combined design strategies. This review emphasized on announcing the importance of advanced geometric layouts like the corrugated absorbers, stepped basins, and hollow fins. These systems introduced an improvement of heat transfer, evaporation and condensation rates. In this context, it was ascertained that trapezoidal absorbers and rotational techniques can upgrade the water productivity by 22.73% and 30.2%, respectively. Specifically, the utilisation of improved materials like graphene-coated absorbers and carbon nanotube Nanofluids can improve the thermal conductivity and energy storage capacity.

The cost and economic analysis conducted in this review has revealed the superiority of the pyramid solar stills that introduced a water production cost between \$0.01–\$0.02 per liter, in addition to having a shorter payback period of between 126 to 185 days, with accommodating low annual CO_2 emissions of 13.7 to 32.2 tons compared to the conventional solar stills systems.

Further enhancements and accompanying challenges

PTSS technology has introduced a notable advancement despite having a number of barriers to attain the optimum efficacy, scalability, and cost-effectiveness. Undoubtedly, addressing these challenges would require interdisciplinary invention and inspired engineering solutions. The current section highlights potential guidelines for future improvements together with the main technological barriers that should be resolved for solar desalination systems to meaningfully upgrade the water productivity and thermal efficiency. The main improvements can be stated in the following directions;

- Upgraded operational duration and enhanced energy storage capacity can be attained via the inclusion of Nano-enhanced PCMs and paraffin wax.
- Hybrid Nanofluids of incorporated TiO_2 and carbon nanotubes should be combined into systems to enhance the thermal conductivity and evaporation rate.
- The hybrid water desalination system of solar stills, photovoltaic-thermal and membrane distillation technologies can meaningfully improve the water productivity.
- Waste heat recovery systems can be integrated into water desalination system to convert thermal losses and upgraded water productivity.
- Inverted pyramid basins and trapezoidal absorbers are effectual solutions to elevate solar energy absorption besides lowering the shadow formation.
- The analysis of environmental and operational data can be conducted using Internet of Things (IoT) sensors and machine learning algorithms in addition to regulating water depth, temperature, and condensation rate to attain the optimum water productivity.
- Automated feed water systems can be used to preserve optimum water levels, while mitigating manual oversight and labor costs.
- Using recycled or low-cost materials (e.g., clay pots, sugarcane bagasse) in the manufacturing process can lower production costs and support circular economy goals.
- Environmental impact can be minimised by reducing energy demands and improving the longevity of materials and components.
- Forced condensation using fans, Peltier devices, and hydrophobic surface coatings can accelerate droplet formation and collection efficiency.
- Investigating fogging mechanisms may improve humidity levels and boost distillate yield.
- Standard testing procedures in different climates, long term reliability experiments of new materials such as Nano-PCMs and combined techno-economic models are suggested to understand scalability and applicability into the real world in a more comprehensive and practical manner. This would deliver a more integrated and practical view of the efficacy of the pyramid solar stills.

However, there is a number of accompanying challenges that should be tackled with as follows;

- PTSSs typically operate at a thermal efficiency between 30–60%, largely dependent on climatic conditions and subject to energy losses.
- Continuous exposure to direct sunlight is necessary, limiting performance during cloudy or low-sunlight conditions.
- Scaling designs for commercial water production while maintaining affordability presents significant engineering and financial hurdles.
- Large-scale installations often require substantial land, which may not be feasible in all regions.
- The efficiency of absorber surfaces and glass covers declines over time due to surface fouling and salt scaling.
- Coastal and saline environments worsen the material degradation and raise the request of periodical maintenance.
- A high-capital cost was ascertained for using the advanced materials like nanoparticles and PCMs.

- The integration of reflectors, PCMs, external condensers can improve the system complication, that may obstruct their widespread distribution.
- Inappropriate dumping of Nanoparticles and the risk of PCM leakage stand for potential ecological hazards.
- The advantages of solar desalination can be undermined as a result to the request of high energy during material production.

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Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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